

Discipline at Work: Wage Premia, Managers, and Shirking in a Large Mexican Retailer

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Andrea Neyra-Nazarrett*

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Abstract

In many jobs, effort takes the form of care: attention a firm cannot measure and a supervisor observes only imperfectly, since a lapse is caught only when someone is watching. What the firm sees is the residue—a loss it can record but not trace to any worker. Inventory shrinkage, merchandise missing from the store, is such a loss, about 1.6 percent of sales for United States retailers, on the order of grocers' net profit margin. I study how wage premia and managerial monitoring jointly deter shrinkage when monitoring is imperfect, using administrative records from a large Mexican retailer with more than 300 stores and roughly 25,000 workers at a time. Leveraging manager rotations, I estimate each manager's effect on shrinkage. The managers who reduce it are not better bookkeepers; they monitor: they work more shifts, including nights and weekends, are more co-present with their workers on the floor, and rework the teams they inherit. On demographic characteristics they are indistinguishable from the rest. Shrinkage also falls with the wage premium. The two reinforce each other: the same arrival reduces shrinkage more where the premium is high, a difference of 8 percentage points. A model of care in which the premium and monitoring are complements can match these facts: where the premium gives the manager's dismissal threat something to confiscate, the same manager accomplishes more.

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1 Introduction

In many jobs, effort takes the form of care: attention a firm cannot measure and a supervisor observes only imperfectly, since a lapse is caught only when someone is watching. The barista who gets an order right, the cook who keeps portions to the recipe, the line worker who lets no defective unit through, the clerk who keeps merchandise from going missing: each supplies care. What the firm sees is the residue: an order remade, food thrown out, a part scrapped, a gap on the inventory count; a cost it can record, but not trace to any particular worker. That is the firm’s problem with care. Because the loss cannot be traced to a particular worker, measures to control it are difficult to implement; a firm cannot pay for performance on care it does not observe, nor charge the loss to whoever caused it. Since the residue is scattered across many small failures, it is rarely recorded systematically at all; where it is, it is large. In United States retail, inventory shrinkage (merchandise that goes missing from stores) reached \$112 billion in 2022, about 1.6 percent of sales, on the order of grocers’ net profit margin (National Retail Federation, 2023; Damodaran, 2026); the most recent global survey puts it at a similar 1.8 percent of sales worldwide (Tyco Retail Solutions and PlanetRetail RNG, 2018). Retailers treat it as a security problem, where they lock merchandise behind glass, expand video surveillance, and increase loss-prevention budgets, while some chains have closed stores because of it (Selyukh, 2024). Despite the cost of lapses in care for firms, it remains unclear which levers a firm holds to sustain care, and there is no systematic way to analyze the problem that could guide a firm in choosing among them.

Two obstacles stand in the way. The first is data. Losses from lapses in care are hard to record; how many orders were remade, how much food a kitchen lost to careless portions or to spoilage, how many units a line scrapped, how much merchandise a store lost. Few firms keep such records across units and over time, and where they do, the number remains an accounting line, never linked to the organizational structure behind it. The second is identification. Even with the data in hand, it is hard to say how much of the loss is the workforce’s doing and how much is attributed to the environment or other parts of the business; a store with high shrinkage may be located in a neighborhood where losses are harder to prevent, and a store with low shrinkage may simply be small. Some machines on an assembly line may be defective, and customers may differ. Care is observed only imperfectly, and a lapse costs the worker something only if it is detected, so telling apart what the firm’s own policies can move from what they cannot requires variation in those policies that is unrelated to the surroundings.

This paper takes on the problem through inventory shrinkage. Shrinkage is linked to care; every step of a retailer’s process runs through workers’ hands (receiving, stocking, handling,

selling), and retail is, at bottom, the custody of merchandise. Where the workforce handles merchandise with care, little goes missing. Where it does not, merchandise goes missing, through neglect and, at the extreme, by the workers' own hand. The same is claimed by evidence produced by retailers: in the industry's own security survey, retailers attribute 56 percent of their shrink to internal causes (their own staff and processes) compared to 36 percent to outside actors (National Retail Federation, 2023), and in my setting, guarding and managing inventory appears in 92 percent of workers' job descriptions. Shrinkage is therefore a direct measure of care, and a way into the wider problem of the cost of lapses in care for firms. I use it to test how two of the levers a firm holds to sustain care function: monitoring, which increases the chance that a lapse is caught, and a wage premium over the market, which increases what the worker loses if it is. The paper analyzes whether monitoring and wage premia incentivize care and, if so, whether they act as complements or substitutes. Using manager rotations across stores, I estimate an AKM model and calculate each manager's effect on inventory shrinkage. I then ask what good managers do by testing the effect of the arrival of a low-fixed-effect (high-type, HT) manager on manager presence in the store and on team management and composition. Good managers are not better bookkeepers; they monitor: they work more shifts, including more night and weekend shifts, are more co-present with their workers on the floor, and rework the teams they inherit. I compare managers with high and low fixed effects and test whether they are statistically different; I find that they are indistinguishable in terms of age, gender, education, and tenure. I then leverage a company policy that sets wages centrally by position and estimate the effect of wage premia on shrinkage. Wage premia reduce shrinkage. Wage premia and monitoring act as complements: the arrival of an HT manager during a high-wage-premium (HWP) period reduces shrinkage by 8 percentage points more than the arrival of an HT manager during a low-wage-premium (LWP) period.

I use administrative records from a large Mexican retailer: more than 300 stores and roughly 25,000 workers in a typical month over 2022–2024. The records measure all three ingredients at once: shrinkage audited weekly in every store; from biometric time clocks, what managers and workers actually do; and each worker's pay relative to her local outside wage. With them, I ask three questions. Does a better monitor reduce shrinkage, and what does she do differently? Does a larger wage premium reduce it? And does the same monitor accomplish more where the premium is high, and are the two complements or substitutes?

I build three panels. The first is an employee-month panel covering close to 100,000 employees over 2022–2024, built from the firm's personnel records. For each employee it records demographics, the position held at each point in time, and wages, from which I measure the premium. I match each employee to Mexico's national labor-force survey (ENOE)

and define her outside option as the average labor income in her cell of gender, education, age, municipality, and quarter; the wage premium is the difference between the wage the firm pays her and that outside wage.

The second is a store-month panel for the firm’s more than 300 stores. It records shrinkage in value and physical units, sales and customer counts from high-frequency scanner data, each store’s format and organizational structure, and its location.

The third is a manager-month panel for the general managers who run each store. It records which manager runs which store in each month along with her demographics, pay, and tenure, and, from the biometric time clocks, when she is physically in the store and the overlap of her schedule with her workers’ shifts.

I take the question to the data with three designs. The first tests the manager. I leverage the firm’s rotation of general managers across stores (a staffing and development policy, not a response to store shrinkage) to estimate a fixed effect for each manager, classify as high-type those whose fixed effects imply less shrinkage, and follow shrinkage around the arrival of a high-type manager at a store. I also use the time clocks, the movement log, and the personnel records to characterize what the managers who reduce shrinkage do in the store. Around the arrival of a high-type manager, shrinkage falls by close to 7 percent, a level I read as validating the classification. The managers who reduce shrinkage are not better bookkeepers; they monitor: they work more shifts, including nights and weekends, are more co-present with their workers on the floor, and rework the teams they inherit, moving workers up a tier and across areas and raising turnover in the short run. They do not add security or supervisory staff, and on demographic characteristics they are indistinguishable from the rest.

The second design tests the premium. Because headquarters sets wages centrally by position, a worker’s premium moves when her local outside wage moves, for reasons unrelated to her store’s shrinkage. I use this variation in a two-way fixed-effects analysis within municipality and time, and I complement it with a statutory shock: Mexico’s minimum-wage hikes compress the premium over the legal floor more at the northern border than in the interior, so I compare border against interior stores of the same format around each hike. The premium disciplines too: a 10 percent lower premium is associated with about 1.1 percent more shrinkage, and around the minimum-wage hikes border shrinkage relative to sales is unchanged in the first months after a hike but rises by about 4 percent in the months that follow.

The third design tests the complementarity, and it asks the question directly: does the same monitor accomplish more where there is more rent to confiscate? I return to the high-type arrivals and split them by the wage environment the manager walks into (whether

the premium in the store’s municipality is above or below that municipality’s own median over the sample), fixed at the month of arrival. After a high-type arrival, shrinkage falls by close to 12 percent where the premium is high against roughly 4 percent where it is low, a gap of eight percentage points. Enforcement is the margin that increases with the premium (suspensions rise only where there is a more valuable rent to confiscate), while the differences in the manager’s other actions are too imprecisely estimated to sign (Figure 8). The two reinforce each other: each instrument is worth most where the other is present. The border hikes close the loop from the other side; when the hikes narrowed the premium, the rise in shrinkage appeared in stores whose manager was seldom on the floor and not in those where she was often present, a pattern the design shows but does not estimate precisely.

I model care by adapting the shirking model of Shapiro and Stiglitz (1984) to the shop floor; the model separates care, observable only when someone is watching, from effort the firm can verify directly, such as attendance. A worker chooses how much care to supply (the hidden effort that keeps merchandise from going missing), and care is costly. The firm pays a fixed wage by position, so the only formal incentive is dismissal: a worker caught neglecting her job loses the premium she earns over her outside option. Whether she is caught depends on the store’s monitoring, which in this setting is the general manager’s presence on the floor. The deterrent is therefore a product: the probability that a lapse is detected times the rent lost upon dismissal.

The product structure delivers the three facts. First, a better monitor disciplines the hidden margin, care: shrinkage falls when a better manager arrives. Second, a larger premium disciplines it too: shrinkage falls as the rent rises. Third, and at the center of the paper, the two are complements; because they enter as a product, and under conditions the model states, a given manager accomplishes more where the premium is high, and a given premium accomplishes more where someone watches the floor. The model also separates margins by observability: attendance is verifiable and detects itself, so the premium moves it everywhere; shrinkage is punished only when detected, so the rent reaches it only through the monitor. Two further propositions sharpen the account: the sanction is the manager’s own act and follows the rent, so she punishes where the job is worth keeping; and a store already watched closely sits at a corner, where a small loss of rent moves nothing.

The paper’s first contribution is the outcome itself. Despite its magnitude, shrinkage has been treated in economics as an accounting line rather than as behavior; to my knowledge, this is the first paper to characterize it systematically and to link it—store by store and manager by manager—to the personnel of the store. Read this way, shrinkage is a measure of care (effort a firm records only as a loss), and the same reading is available wherever a firm audits such a loss.

Second, the paper contributes to the evidence that pay premia sustain discipline within firms. The classic reference is the interplant test of Cappelli and Chauvin (1991), who relate the premium a manufacturer pays across plants to disciplinary dismissals; the closest recent work is Emanuel and Harrington (2024), who show that a voluntary firm-wide minimum wage halves worker departures, reduces absenteeism, and raises warehouse productivity. The present study adds variation: pay is set centrally while local outside wages move, and two statutory minimum-wage schedules move the premium by law, so the premium moves both with local labor markets and by statute, across over three hundred stores, against the same audited outcome.

Third, and closest to this paper, Coviello et al. (2022) establish a wage-by-monitoring complementarity: a minimum-wage increase raises the productivity of bound workers, and by more where monitoring is intensive. This paper differs in how monitoring is measured. Their proxy is a staffing quantity, the store-month supervisor-to-worker ratio, which, as the authors note, “captures the extensive margin of monitoring but not the intensive margin (supervisor effort).” Here, the monitor is an individual manager whose arrivals and departures I observe. Rotation converts monitoring into discrete, dated, within-store events rather than gradual changes in a staffing ratio; the time-clock records measure the manager’s presence directly (her hours, her store, and her overlap with her workers), the intensive margin a staffing ratio cannot capture; and the manager becomes an object of direct interest, since how managers complement a firm’s other incentive instruments is a central question of personnel economics.

Finally, the paper adds an outcome to the literature that recovers manager effects from rotation and asks what good managers do (Bertrand and Schoar, 2003; Lazear et al., 2015; Minni, 2026; Adhvaryu et al., 2026). The novelty lies in the margin and its interpretation: the manager’s effect is estimated on a margin a supervisor observes only imperfectly, traced to concrete managerial conduct, and shown to depend on the wage environment in which the manager operates.

2 Data and Institutional Background

2.1 The Firm and the Data

I partner with a large Mexican retailer with a nationwide presence. The firm operates more than three hundred stores across the country and employs around twenty-five thousand workers in any given month; the panels cover roughly one hundred thousand distinct workers over the full period. Central headquarters standardizes stores tightly within format, setting the physical layout, the wage attached to each position, the human-resources and personnel policies, and the operating routines that govern each location. Table 1 summarizes the firm’s scale and the structure of the data, which span January 2022 through June 2024.

Table 1: Company summary

Category	Description
Country	Mexico
Sector	Retail
Number of workers	around 100,000
Number of stores	300+
Stores outside border zone	225+
Stores inside border zone	75+
Period	January 2022 – June 2024

Data access provided through a research partnership. All analyses use anonymized records; store identifiers and exact locations are confidential.

Using the firm’s administrative records and the ENOE, I build three linked panels. The first is the employee-month panel, from the personnel and scheduling system and the time clocks: demographics, position, and the description, tasks, and actions of each job; wage and wage premium; shifts and hours, with each entry and break stamped to the minute; absences with their reason; suspensions; and every move—promotion, transfer, separation—with the reason it occurred. The biometric time clock is the same instrument for everyone in the store: it stamps the start and the end of every worked shift to the minute, for frontline workers and general managers alike, so the records place each person in the store day by day—nights and weekends included—and any two employees’ schedules can be laid over one another. The personnel-movement record behind these variables is a log of events, not a snapshot: it registers every hire, every separation with its recorded cause—quits, dismissals, and severances enter separately—every within-store reassignment across positions, tiers, and occupational areas, and every promotion and demotion, each stamped with the month it occurred. The same system carries the disciplinary record: every formal suspension is logged

with its date, so the sanctions applied in a store can be counted month by month. The second is the store-month panel: scanner-level sales aggregated to the month, the weekly inventory audit’s shrinkage in value and physical units, format, and location. The third is the manager-store panel, from the same personnel system, which follows each general manager across stores over time.

2.2 Store Scale and Structure

Table 2 reports the distribution of sales and headcount at the store level. Each store has three tiers: a managerial layer of one or more general managers (up to three in the largest formats), a middle tier of area sub-managers and supervisors, and the general workforce.

Table 2: Store sales and activity (store-month)

Variable	Mean	Median	Min	Max
Sales (USD)	682,726	550,828	136,785	2,087,354
Sales (units)	426,469	347,440	95,543	1,197,007
Clients (count)	47,123	42,156	3,608	118,945
Workers per store	67.3	54	1	262

Store-month observations (more than 300 stores, January 2022–June 2024). Monetary values are in USD, converted at MXN 18.5/USD, the approximate average interbank rate over the period. Inventory shrinkage is reported separately in Table 4.

2.3 Store Managers

The general manager hires and dismisses workers, assigns and reassigns them across occupational areas, promotes and moves them within the store, sets the schedule, and oversees inventory. Because pay is fixed centrally by position, none of these levers runs through compensation: the manager can neither raise nor cut a worker’s wage, so her influence lies in selection, allocation, scheduling, and supervision.

2.4 Wage Premia and Shirking Measures

Two constructs anchor the analysis: the wage premium, which measures the rent the firm pays over the outside option, and a set of shirking measures, of which inventory shrinkage is the central one.

Wage premia Following the efficiency-wage logic of Cappelli and Chauvin (1991), I define the wage premium as the difference between a worker’s wage inside the firm and the wage

the same worker would command outside it,

$$W_{it}^p = w_{it} - w_{it}^m, \quad (1)$$

where w_{it} is worker i 's firm wage in month t and w_{it}^m is her outside wage. I measure the outside wage by matching each worker to a reference profile in the ENOE, the national labor-force survey administered by INEGI, which covers both formal and informal workers: the average labor income of employed respondents in her full cell—same gender, education, age, municipality, and quarter—which accounts for about 86 percent of worker-months. Where the full cell is empty, the match relaxes one dimension at a time, education last, and never drops the quarter or the state; 99 percent of worker-months obtain an outside wage.¹ Table 3 reports the firm wage, the outside wage, and the premium in levels and as a multiple of it.

Table 3: Wages and the wage premium (worker-month)

	Mean	Median
Firm wage (USD)	476	428
Outside wage, ENOE (USD)	382	364
Wage premium (USD)	93	69
Wage premium (multiple)	1.33	1.20

Worker-month observations. The firm wage is monthly contractual pay; the outside wage is the ENOE cell wage matched on gender, education, age, municipality, and quarter (the cascade for empty cells never drops the quarter or the state). The wage premium in levels is the difference between the two; the multiple is the ratio of the firm wage to the outside wage. USD at MXN 18.5/USD.

Shirking To gauge shirking and effort I use four personnel outcomes: turnover, the monthly rate at which workers leave the store (the records carry the reason of each separation); absenteeism, in days and as an indicator for any absence, separating non-justified from justified; and, from the scheduling system, monthly shifts and daily worked hours.

As in other large retailers, tenure is short and absence and turnover are high: median tenure is 26 months. The paper's central measure is inventory shrinkage—the audited value of merchandise the firm records as owned but cannot find on the shelf.² The firm's weekly inventory audit measures it in both value and physical units: it compares the book stock the

¹Relaxing gender, then age, then municipality accounts for a further 0.7, 0.5, and 1.2 percent of matches (so about one percent are matched at the state rather than the municipality level); education-accounts for the remaining 12 percent.

²Merchandise that is spoiled or damaged but still identifiable is logged in a separate register and is not counted as shrinkage, so the measure captures stock that is simply missing.

firm's system expects a store to hold against a physical count of what is on the shelf, and records the gap. The firm counts at three frequencies—a daily reconciliation of book against physical stock and fuller counts each week and each month—so losses are caught close to when they occur; at the close of each month it reconciles the flow identity (inflows = sales + shrinkage), so recorded losses tie out against measured stock movement. Who takes the count changes from week to week; it is not a fixed person. The audit is of the store: the register records the gap for the store as a whole and charges it to no worker. In this firm shrinkage averages 2.53 percent of sales.

Interpreting shrinkage as shirking rests on an assumption about where the losses come from. Shrinkage responds to two behaviors of the workforce: the attention workers devote to preventing losses on the floor—watching merchandise, handling it with care, flagging problems as they appear—and the losses workers themselves cause. Care moves both margins in the same direction: high effort prevents shrinkage, while low effort stops preventing it or, at the extreme, causes it. The data support this reading: store shrinkage covaries most with the workforce and organization—stores whose workforces are less experienced, less stable, and more often absent without justification carry more shrinkage—while customer traffic enters weakly and local robbery, homicide, and perceived insecurity cluster near zero (Appendix L). Table 4 reports summary statistics for shrinkage and the remaining shirking measures at the store and worker levels.

Table 4: Shirking measures at the store and worker levels

Variable	Mean	Median	Min	Max
Store level (store-month)				
<i>Inventory shrinkage</i>				
Shrinkage-to-sales (%)	2.53	2.45	0.10	10.52
Shrinkage (USD)	16,931	12,450	807	67,329
Shrinkage (units)	6,440	5,012	47	21,497
Shrinkage per worker (USD)	242	230	16	1,219
<i>Absence (store total, days per month)</i>				
Absence days (total)	126.3	101	0	626
Non-justified absence days	58.2	46	0	313
<i>Turnover (% of workforce per month)</i>				
Turnover rate (total)	6.20	5.48	0	31.25
Worker level (worker-month)				
<i>Absence (days per month)</i>				
Absence days (total)	1.88	0	0	31
Non-justified absence days	0.86	0	0	28
<i>Absence (indicator)</i>				
Any absence (= 1)	0.462	0	0	1
Any non-justified (= 1)	0.361	0	0	1
<i>Effort</i>				
Monthly shifts	21.33	25	1	56
Daily worked hours	8.26	7.92	0	24

Store statistics are computed at the store-month level (more than 300 stores); worker statistics at the worker-month level, January 2022 through June 2024. USD at MXN 18.5/USD. At the store level, absence days are totaled across the store’s workers, and turnover rates are separations per hundred workers. At the worker level, absence days are per-worker monthly counts and the absence indicators equal one if the worker had at least one absence of that type. *Non-justified* absence is absence without an approved justification; vacation days are reported separately and excluded here. Effort margins (shifts, hours) are recorded by the scheduling system.

3 Theoretical Framework

This section formalizes how wage premia and managerial monitoring jointly deter shirking when monitoring is imperfect. I model the effort choice of a worker whose effort is care—attention a supervisor observes only imperfectly—and whose only incentive is the threat of losing a job that pays a premium over the local market wage. The model adapts Shapiro and Stiglitz (1984) and is the pure efficiency wage case of Coviello et al. (2022): fixed salaries, no pay-for-performance, an outside option equal to the market wage rather than unemployment, and dismissal as the firm’s only formal incentive. The expected penalty for shirking is a *product*: the probability that a lapse is detected, which the manager sets, times the rent the worker forfeits if sanctioned, which the premium sets. The results follow the order of the paper—the monitor, the premium, their product, and who enforces it—and each proposition names the section that tests it.

3.1 Environment

Worker i of type x_i is employed at store j under manager m . Each period she chooses a *care level* $e_i \in [0, 1]$ —the hidden effort that keeps merchandise from going missing, including keeping time on the floor once she is there—and an *absence level* $a_i \in [0, 1]$, the share of assigned shifts she shows up for. Care has consequences only if someone detects the losses it produces; absence is recorded by the time-clock and detected mechanically, whoever runs the store. Store shrinkage is $S_j = f(1 - \bar{e}_j)$, with $\bar{e}_j$ average care and f strictly increasing. Costs are $c(x_i, e)$, with $c_e > 0$ and $c_{ee} > 0$, and $\kappa(a)$, increasing and convex.

The worker receives a fixed wage

$$w_i = w^m + W_i^p, \quad (2)$$

where w^m is the local market wage she can obtain immediately if she leaves³ and $W_i^p \geq 0$ is the *wage premium* (Section 2, Equation 1). The wage does not depend on realized shrinkage. The model is stationary: the propositions are comparative statics in the standing premium, and each worker-month observation maps to the model evaluated at its current premium, treated as permanent.

The job ends through two hazards, one per margin. On the care margin, manager m monitors with intensity $\mu_m \geq 0$; a lapse by a worker exerting care e is detected with probability $q(\mu_m)h(e)$ and, in the baseline, sanctioned automatically, so she keeps her job

³Immediate reemployment is a normalization. With a search spell in informal work at $w_u < w^m$ and reemployment at rate λ , everything below holds with the premium augmented by $\delta \equiv (w^m - w_u)r/(\lambda + r)$, about three percent of the median premium under ENOE spells of one quarter.

with probability

$$\pi(e; \mu_m) = 1 - b - q(\mu_m) h(e), \quad b \in (0, 1), \quad q > 0, \quad q' > 0, \quad h \geq 0, \quad h' < 0, \quad h'' \geq 0, \quad (3)$$

with b an exogenous separation rate, q the detection rate the manager generates, and h the worker's exposure to detection. Hence

$$\pi_e = q(\mu_m)(-h'(e)) > 0, \quad \pi_{e\mu} = q'(\mu_m)(-h'(e)) > 0 : \quad (4)$$

more care lowers the dismissal risk, and closer monitoring rewards care more strongly. The parametric form is not innocuous (Section 3.5). On the absence margin, retention is $\pi^A(a)$ with $\pi^A > 0$ and no μ_m . Treating the two hazards as mutually exclusive within a period,

$$\Pi(e, a; \mu_m) = \pi(e; \mu_m) + \pi^A(a) - 1, \quad (5)$$

which must lie in $[0, 1]$. The additive form is substantive: a multiplicative one would place μ_m inside the absence condition.

3.2 The Worker's Problem

Time is discrete. Let $V^E(x_i)$ be the value of employment and V^A that of the outside option, with flow value $u^A = \frac{r}{1+r} V^A = w^m$ since the worker can always find work at w^m . Writing the employment Bellman equation and subtracting V^A , the *employment surplus* $V(x_i) \equiv V^E(x_i) - V^A$ satisfies

$$V(x_i) = \max_{e, a \in [0, 1]} \left[W_i^p - c(x_i, e) - \kappa(a) + \frac{\Pi(e, a; \mu_m)}{1 + r} V(x_i) \right]. \quad (6)$$

The market wage drops out: only the premium determines incentives. At an interior optimum the first-order conditions are

$$c_e(x_i, e^*) = \frac{\pi_e(e^*; \mu_m)}{1 + r} V(x_i), \quad (7)$$

$$\kappa'(a^*) = \frac{\pi^A(a^*)}{1 + r} V(x_i), \quad (8)$$

and the surplus at the optimum is

$$V(x_i) = \frac{(1 + r)[W_i^p - c(x_i, e^*) - \kappa(a^*)]}{1 + r - \Pi(e^*, a^*; \mu_m)}. \quad (9)$$

Both conditions price the surplus the worker stands to lose; only the care condition runs through the manager's monitoring.

Definition 1 (Stationary partial equilibrium of the worker). *Given (x_i, W_i^p, μ_m) , a triple (V, e^*, a^*) such that (a) $0 \leq \Pi(e, a; \mu_m) \leq 1$ for all (e, a) ; (b) (e^*, a^*) maximizes the right-hand side of (6) given V , with Kuhn–Tucker conditions $g_e \equiv -c_e + \pi_e V/(1+r) \leq 0$ if $e^* = 0$, $= 0$ if interior, ≥ 0 if $e^* = 1$, and likewise $g_a \equiv -\kappa' + \pi^A V/(1+r)$; (c) V is the fixed point (9); (d) $V \geq 0$.*

Nothing here is a firm's or a market's equilibrium: (W_i^p, μ_m) are exogenous. Write $\rho \equiv 1 + r - \Pi(e^*, a^*; \mu_m)$, $D \equiv c_{ee} - \pi_{ee} V/(1+r)$, $D_A \equiv \kappa'' - \pi^{AA} V/(1+r)$, and let $R = W \times [\mu_{LT}, \mu_{HT}]$ be a rectangle of premiums and monitoring intensities on which (e^*, a^*) is interior, with c and h three times and κ, π^A twice continuously differentiable.

Lemma 1. *Under (3) and $D_A > 0$: (i) V exists and is unique, the care problem is single-peaked, and $D > 0$; (ii) the care condition is $M(e^*) = q(\mu_m) V/(1+r)$ with $M(e) \equiv c_e(x_i, e)/(-h'(e))$ and $M' > 0$, so care is a strictly increasing transform of the expected-penalty product; (iii) V is strictly increasing in W_i^p , and $V > 0$ at an interior optimum requires $W_i^p > c(x_i, e^*) + \kappa(a^*)$.*

Proofs are in Appendix A.

3.3 Linking the Model to the Setting

The monitor. The argument of (7) is the monitoring the worker actually faces in store j in period t , $\mu_{mjt} = \bar{\mu}_m + u_{mjt}$, with $\bar{\mu}_m$ portable. The arrival designs classify managers by their AKM shrinkage fixed effect (Section 4), high-type (HT) below the median of their connected set; Section 5 grounds the label in conduct: high-type managers work more shifts, overlap more of their workers' hours, and their workers keep better time while they are on the floor. Monitoring is not an apparatus the manager purchases but her presence on the floor, which proxies μ_{mjt} itself; nothing in her problem re-optimizes it against the local rent, so her actions should not vary with the premium while the workers' response does (Section 7.3). Where label and realized presence disagree as moderators (Section 8.1), the model sides with the realized value.

The premium. In Shapiro and Stiglitz (1984) the punishment for shirking is unemployment; here the market always offers w^m , what dismissal destroys is the premium, and (6) makes the premium the whole of the incentive. Workers earn on average about 1.33 times their outside wage (Section 2). Because the contract wage is set centrally by position and

moves rarely, movements in the local outside wage pass through one-for-one to the premium, the variation the two-way fixed-effects design of Section 6 exploits.⁴ A statutory minimum-wage hike (Section 8) moves the premium in two phases with opposite signs: on impact it raises the contract wage of the workers the floor covers, the case Coviello et al. (2022) study; as the local outside wage catches up against a scale that does not move, the premium compresses, the case they set aside as demotivating. Proposition 2 signs both; the stationary model does not date the second.

The no-shirking condition. In the binary case of Shapiro and Stiglitz (1984)— $e \in \{0, \bar{e}\}$, absence suppressed, retention $1 - b$ with effort and $1 - b - q$ without—the surplus of a worker who exerts effort is $V_H = (1 + r)(W_i^p - \bar{e})/(r + b)$, a one-period deviation saves $\bar{e}$ and exposes V_H to detection risk q , so effort is optimal if and only if

$$W_i^p \geq \bar{e} + \frac{(r + b)\bar{e}}{q} \equiv \widehat{wp}, \quad (10)$$

the effort cost, which a rent gross of effort must cover, plus the no-shirking mark-up. Read as the firm’s menu, the condition makes the two instruments *substitutes*: the same care can be sustained with more monitoring and less premium. That is the null the arrival design of Section 7 rejects in levels.

3.4 Results

Proposition 1 (The monitor disciplines the hidden margin). *Let (e^*, a^*) be interior. Under higher monitoring intensity μ_m :*

- (i) *Care is higher at any given premium, $\partial e^*/\partial \mu_m > 0$, and shrinkage lower.*
- (ii) *Absence does not fall. Each margin’s response to the manager is its response to the premium rescaled by a detection stake,*

$$\begin{aligned} \frac{\partial a^*}{\partial \mu} &= -\frac{q'(\mu) h(e^*) V}{1 + r} \cdot \frac{\partial a^*}{\partial W^p} \leq 0, \\ \frac{\partial e^*}{\partial \mu} &= \frac{q'(\mu) V}{1 + r} \cdot \frac{r + b + 1 - \pi^A(a^*)}{q(\mu)} \cdot \frac{\partial e^*}{\partial W^p} > 0, \end{aligned} \quad (11)$$

so a stricter monitor moves absence only through the residual exposure $h(e^)$ of a*

⁴The treatment of monitoring parallels Coviello et al. (2022): detection depends on effort when monitored, termination follows detection, monitoring is exogenous. Their μ is the share of workers monitored in a store; here μ_m is the intensity embodied in the manager.

compliant worker, zero in the benchmark $h(e^*) = 0$, while her effect on care runs through the slope $-h'(e^*)$ and is bounded away from zero.

- (iii) With an unanticipated shock $\varepsilon \sim G$ to the outside option, realized after actions, quits rise with μ_m in proportion to $g(V) q'(\mu) h(e^*) V / \rho$: only among workers who still carry exposure. The dismissal rate of a compliant worker, $q(\mu) h(e^*)$, stays near zero.

Part (i) uses the form (3), not only the signs (4): a stricter monitor also trims the surplus of a worker with exposure, and under (3) the deterrence channel dominates.⁵ Sections 4 and 5 test the proposition: shrinkage falls when a high-type manager arrives and climbs back when she leaves; turnover rises; and the test of (ii) is the arrival design on non-justified absence (Section 7, Appendix H), not the comovement of manager effects across outcomes, which validates the label and is silent on the mechanism.

Proposition 2 (The premium disciplines both margins). *At an interior optimum:*

- (i) Care rises, $\partial e^* / \partial W^p = \pi_e / (\rho D) > 0$, and shrinkage falls.
- (ii) Absence falls, $\partial a^* / \partial W^p = \pi^{A'} / (\rho D_A) > 0$, and time on the job rises.
- (iii) The dismissal hazard $1 - \Pi$ falls, and with the shock of Proposition 1(iii) quits fall as well.

Sections 6 and 8 test it: shrinkage, absence, and turnover fall and shifts and hours rise with the premium, and the statutory compression raises shrinkage. Two assumptions sign what follows.

Assumption 1 (Compliant-care detection). (a) The care hazard is (3). (b) At every optimum in R , the residual exposure of compliant care is small relative to its slope:

$$h(e^*) \pi^{A'}(a^*) \kappa'(a^*) D < (-h'(e^*)) c_e(x_i, e^*) (r + b + 1 - \pi^A(a^*)) D_A. \quad (12)$$

Assumption 2 (Curvature regularity). $M(e) = c_e(x_i, e) / (-h'(e))$ is log-concave, $(\log M)'' \leq 0$.

Part (b) is “ $h \approx 0$ at compliant care” as an inequality; the benchmark $h(e^*) = 0$ satisfies it strictly. Assumption 2 disciplines the third-order terms the cross-partial of an optimum drags in and holds whenever c_e and $-h'$ have constant elasticities.

⁵Under $\pi_e \geq 0$, $\pi_{e\mu} \geq 0$ alone, $\partial e^* / \partial \mu = \frac{V}{(1+r)D} [\pi_{e\mu} + \pi_e \pi_\mu / \rho]$ carries the unrestricted sign of π_μ : with $c = e^2/2$, $r = 0.05$, $W^p = 0.05$, $\pi = 0.8 + \mu e - 0.4\mu^2$, at $\mu = 0.9$ the solution is $e^* \approx 0.084$, $\pi^* \approx 0.55$, and $\partial e^* / \partial \mu \approx -0.014$. The form (3) is a parametric member of the monitoring order of Coviello et al. (2022, Definition 2), which rules such cases out.

Proposition 3 (The premium and the monitor are complements). *Let $e^*(W^p, \mu)$ solve (7) on R under Assumption 1(a). Then:*

(i) (**One product does all the work.**) *Optimal care satisfies*

$$M(e^*) = q(\mu) \frac{V}{1+r}, \quad M' > 0, \quad (13)$$

with slopes

$$\frac{\partial e^*}{\partial W^p} = \frac{\pi_e(e^*; \mu)}{\rho D} > 0 \quad \text{and} \quad \frac{\partial e^*}{\partial \mu} = \frac{q'(\mu)(-h'(e^*))(r+b+1-\pi^A(a^*))V}{(1+r)\rho D} > 0. \quad (14)$$

(ii) (**Local complementarity.**) *Under Assumptions 1–2,*

$$\frac{\partial^2 e^*}{\partial W^p \partial \mu} > 0 \quad \text{on } R. \quad (15)$$

(iii) (**The premium amplifies the monitor.**) *For every $W^p \in W$, $\Delta_\mu e^*(W^p) \equiv e^*(W^p, \mu_{HT}) - e^*(W^p, \mu_{LT})$ is strictly positive and strictly increasing in W^p : for $W_L^p < W_H^p$ in W ,*

$$\Delta_\mu e^*(W_H^p) - \Delta_\mu e^*(W_L^p) = \int_{W_L^p}^{W_H^p} \int_{\mu_{LT}}^{\mu_{HT}} \frac{\partial^2 e^*}{\partial W^p \partial \mu} d\mu dW^p > 0. \quad (16)$$

Propositions 1(i), 2(i), and 3 are three readings of (13): the derivative of the product in the detection rate, in the rent, and their cross-partial. The premium's proportional grip on the surplus is $\partial \log V / \partial W^p = 1/(W^p - c - \kappa)$, and a stricter monitor raises the cost of the actions the worker chooses, so the same peso moves the surplus by more in proportion; the sign is established under Assumptions 1(b) and 2, not by the product alone. Part (iii) is what Section 7 tests cell by cell—the reinforcement Coviello et al. (2022) find from the opposite direction.

Enforcement. In the store the sanction is the manager's act—a suspension or a dismissal she must file, argue, and staff around—and it costs her.⁶ What it buys her is deterrence, which runs only through what the sanction confiscates: a worker with no rent to lose is indifferent to losing the job. Suppose she sanctions a detected lapse if and only if the rent it

⁶A dismissal confiscates the rent V ; a suspension without pay confiscates a share of it. The threshold logic is the same, and suspensions are the sanction the records observe (Section 7.3).

confiscates is at least $\bar{v} > 0$. Writing $s \equiv \mathbf{1}[V \geq \bar{v}]$, with V the surplus enforcement sustains, the care hazard becomes $\pi = 1 - b - s q(\mu) h(e)$: the detection technology is the manager's presence, but the threat behind it is exercised only where there is rent to confiscate.⁷

Proposition 4 (Enforcement follows the rent). *Under the sanction rule and Assumption 1(a):*

- (i) (**Credibility threshold.**) *There is a premium threshold $\bar{W}(\mu)$, defined by $V(\bar{W}, \mu) = \bar{v}$, such that enforcement is credible if and only if $W^p \geq \bar{W}(\mu)$. Below it $\pi_e = 0$: care sits at its lower corner and responds neither to the premium nor to the monitor.*
- (ii) (**Sanctions on impact.**) *When a stricter monitor replaces a laxer one, $\mu_L \rightarrow \mu_H$, at the care level workers chose under the old monitor the sanction rate rises by $[q(\mu_H) - q(\mu_L)] h(e_L^*)$ where enforcement is credible under the new monitor, $W^p \geq \bar{W}(\mu_H)$, and by nothing where it is not.*
- (iii) (**Extensive margin.**) *Above the threshold Proposition 3 holds unchanged; across it, the arrival effect on care rises from zero to $\Delta_\mu e^*(W^p) > 0$. The share of a store's workers above the threshold rises with the store's premium, so the store-level arrival effect rises with the premium through both margins.*

Proposition 4 is the mechanism Section 7.3 tests. The arrival of the same manager should show the same presence in every wage environment, more sanctions where the premium is high and none where it is low, and a larger fall in shrinkage where the premium is high. The signed object is the differential in sanctions on impact; the stationary sanction rate $q(\mu_H) h(e_H^*)$ is not signed. The same rule explains why no premium rescues a store from a weak monitor: with $q(\mu)$ near zero the sanction rate is near zero at any rent (Section 7, low-type panel).

Corollary 1 (Aggregation). *Let store j employ N_j workers with premiums $W_{ij}^p = \bar{W}^p_j + \eta_i$, the joint distribution of (x_i, η_i) fixed, a common μ_j , every (W_{ij}^p, μ_j) in R , and $S_j = \alpha_j + \frac{1}{N_j} \sum_i \ell(1 - e_{ij}^*) > 0$ with ℓ linear and increasing. Under the hypotheses of Proposition 3, $\partial S_j / \partial \bar{W}^p_j < 0$, $\partial S_j / \partial \mu_j < 0$, $\partial^2 S_j / \partial \bar{W}^p_j \partial \mu_j < 0$, and the same signs hold for $\log S_j$.*

With convex losses, $f'' > 0$, the corollary requires $f'' \bar{e}_{W^p} \bar{e}_\mu < f' \bar{e}_{W^p \mu}$; it also holds composition fixed, which the team evidence of Section 5 licenses.

⁷The rule is a commitment evaluated at the surplus under enforcement. Evaluated lapse by lapse it leaves a band of premiums, between the surplus a non-enforcing manager's workers hold and $\bar{v}$, where the only consistent policy sanctions with some probability and the threshold of part (i) becomes a ramp.

Corners. Section 8.1 reads a pattern the interior results do not predict: a marginal compression of the premium raises shrinkage only where the manager is not on the floor. On R , by (15), an interior compression removes *more* care where monitoring is high. The insured store is a corner. By Definition 1(b) and Lemma 1(ii), care is at full compliance, $e^* = 1$, if and only if

$$\begin{aligned}\Gamma(W^p, \mu) &\equiv \frac{q(\mu) V^{(1)}(W^p, \mu)}{1+r} - M(1) \geq 0, \\ V^{(1)}(W^p, \mu) &\equiv \frac{(1+r)[W^p - c(x_i, 1) - \kappa(a^*)]}{r+b+q(\mu)h(1)+1-\pi^A(a^*)},\end{aligned}\tag{17}$$

with $V^{(1)}$ the surplus at full care and a^* solving (8) at $V = V^{(1)}$.

Proposition 5 (Saturation and marginal insurance). *Under Assumption 1(a), with $D_A > 0$ and $V^{(1)} > 0$:*

- (i) Γ is strictly increasing in W^p and in μ , so there is a threshold $\bar{\mu}(W^p)$, decreasing in W^p , with $e^*(W^p, \mu) = 1$ if and only if $\mu \geq \bar{\mu}(W^p)$.
- (ii) (**Marginal insurance.**) Fix W^p and $\mu_L < \bar{\mu}(W^p) < \mu_H$. For every sufficiently small compression $dW^p < 0$,

$$\begin{aligned}e^*(W^p + dW^p, \mu_H) - e^*(W^p, \mu_H) &= 0, \\ e^*(W^p + dW^p, \mu_L) - e^*(W^p, \mu_L) &= \frac{\pi_e}{\rho D} dW^p + o(dW^p) < 0.\end{aligned}\tag{18}$$

- (iii) If instead both (W^p, μ_L) and (W^p, μ_H) lie in R under Assumptions 1–2, the ordering reverses: the same compression removes more care under the strict monitor.

Section 8.1 tests Proposition 5. It is not the marginal form of the complementarity—part (iii) orders the stores the other way—but a statement about where each store sits relative to the corner: a compression damages the unwatched store because its care margin is interior and spares the watched one because a small loss of rent leaves the slack Γ positive. Part (i) says which stores should be saturated, those with a high premium *and* a strict monitor, and a compression large enough to push Γ below zero would end the insurance. The same logic on the absence margin, $g_a(1) \geq 0$, insures the attendance of a worker at full compliance.

3.5 Discussion

Workers are risk neutral throughout. Appendix A.2 records where the data refine the model: monitoring is an attribute embodied in the manager rather than a technology the firm

buys; the complementarity lives in the worker’s first-order condition rather than in the manager’s conduct, which barely responds to the premium; and shirking splits into a care margin that must be detected and an attendance margin that records itself, so that the level complementarity and the marginal insurance of the watched store are two comparative statics of one condition. Three comments.

The detection technology is load-bearing. The form (3) delivers concavity in care by construction and carries the property the two-margin results rest on: a compliant worker has little to fear from a watchful manager, so monitoring changes the return to care, not the survival odds of the compliant; without it the monitoring comparative static can reverse. Assumption 1(b) is stated at the optimum: by Equation (32) of Appendix A.1 it is equivalent to a positive induced-retention response, $\beta \equiv \pi_e \partial e^* / \partial \mu + \pi^{A'} \partial a^* / \partial \mu > 0$, which under linear exposure $h = h_0(1 - e)$ holds whenever compliant care is close to full or the absence hazard is flat at the optimum. Exact invariance of absence is a limit, not a specification: $h \equiv 0$ would erase the care margin too, so the model delivers the bound in (11), which vanishes at fixed slope only as $e^* \rightarrow 1$, the saturation region of Proposition 5. $D_A > 0$ is assumed, with sufficient conditions parallel to Assumptions 1–2 of Coviello et al. (2022).

What the stationary model does not predict. Timing: the model compares steady states and does not date the compression’s lag (Sections 8 and 8.1); a partial-adjustment extension, $e_{t+1} = (1 - \lambda)e_t + \lambda e^*(W_t^p, \mu_t)$, delivers it, and its impact period is where Proposition 4(ii) lives. Team production: the hazard $q(\mu)h(e_i)$ assumes detection identifies individual negligence; were losses attributable only to the store, care would be a public good within the team. Corollary 1 holds N_j fixed.

The low-type pattern. A low-type arrival raises shrinkage in both wage environments with no significant difference between them (Section 7). The model predicts the sign, by (14), not the absence of scaling: on the interior, (15) makes the damage from a weak monitor larger where the premium is high. The flat pattern is descriptive, consistent with imprecision or with the low-premium store sitting at the lower corner under the weak monitor. What the model does license is the weaker statement: a premium disciplines the shrinkage margin only through a monitor who enforces (Proposition 4).

4 Manager Effects on Shrinkage: AKM Decomposition

I study the general manager’s effect on inventory shrinkage using the firm’s rotation of managers across stores. I proceed in three steps. First, I estimate a two-way fixed-effects model following Abowd et al. (1999) (AKM), leveraging manager rotations that are unrelated to store shrinkage. Second, I classify managers as high- or low-type by their estimated fixed effects. Third, I take the classification to the arrival-and-departure design of Section 5, which tests it against store transitions and asks what high-type managers do.

The model is

$$Y_{jt} = \theta_{m(j,t)} + \psi_j + \delta_t + \varepsilon_{jt}, \quad (19)$$

where Y_{jt} is log shrinkage value in store j and month t , $\theta_{m(j,t)}$ is a fixed effect for the general manager running the store, ψ_j absorbs permanent store characteristics, and δ_t absorbs seasonality and firm-wide shocks. Following Card et al. (2013), the error decomposes as

$$\varepsilon_{jt} = \eta_{m(j,t),j} + \xi_{jt} + \nu_{jt}, \quad (20)$$

a manager–store match component, a store component, and transitory noise. Identification requires that manager–store assignment be mean-independent of ε_{jt} : managers may sort on the permanent components, which the fixed effects absorb, but not on the match or on transitory shocks (Appendix B states the moment conditions).

Three facts show that manager assignment is unrelated to store shrinkage, in levels and in the match-specific component. First, although the firm audits shrinkage weekly, it is not a key performance indicator in manager assignment: human resources does not track it systematically, and, by the firm’s own descriptions of manager reassignments, placement is not directed by the shrinkage of the destination store—in particular, and from conversations with the firm, the firm does not send its best managers to the stores with the most shrinkage. Second, the movement records themselves show coordinated rotation rather than targeted placement. Table 14 (Appendix B) classifies the roughly 520 between-store moves: nearly half are links in a same-month chain; the origin store receives another moving manager the month its own manager departs; one in five is a direct swap between two stores, and one in three fills a vacant store; and the moves cluster in waves—a handful of months absorb the largest share—consistent with coordinated administrative rotation rather than case-by-case reaction. The distance the moves travel in shrinkage is zero: the destination’s pre-move shrinkage exceeds the origin’s shrinkage by 0.026 log points ($p = 0.27$).

Third, the moves pass the two direct tests of exogenous mobility. The fixed effects, and everything built on them, rest on the assumption that the firm does not assign managers

to stores on the basis of shocks to shrinkage; the rotation policy makes that plausible, and Appendix Figure 12 tests it directly, with the event-study test of Card et al. (2013). I rank stores by quartile of their average shrinkage, isolate the managers who move across quartiles, and follow each manager across her move: the outcome at every date is the shrinkage of whatever store she is running, so at the move the store changes and the manager does not. If assignment were unrelated to the error term, a mover’s outcome should jump to the destination store’s level on impact—the jump measures the store, not the manager—with no differential trend before the move and with upward and downward moves mirroring each other. The figure shows this pattern: movers from bottom- to top-quartile stores step up to the destination level on impact and the reverse moves fall symmetrically; within-quartile moves are flat, so the step is not an artifact of moving itself; and trajectories are flat within origin quartile before the move, so movers were not selected on a diverging shrinkage path. The stores, not the managers, carry the level differences the figure displays, which is what the decomposition assumes when it separates the two.

The second test uses the symmetry that mean-independence imposes on the match component of Equation (20). If managers moved to stores where their particular match is favorable, moving up and moving down the store-quality ladder would both look advantageous, and the gains from upward moves would exceed the losses from downward ones. Appendix Figure 13 plots the average residual change for each upward quartile pair against the corresponding downward pair; symmetry implies the -45 -degree line. The points cluster around that line, and no pair deviates in the direction implied by match-based sorting: what a manager gains by moving up a quartile, another loses by moving down it. Together with Figure 12, this is what allows the fixed effect to be read as the manager’s own contribution rather than as the effect of the store to which she was assigned.

I also address limited-mobility bias. A majority of general managers are movers: 55.0 percent run at least two different stores in the sample, 25.4 percent run three or more, and 91.7 percent of stores receive at least one mover (Table 13, Appendix B)—high relative to the matched employer–employee panels on which AKM methods were developed, where mover shares of 12, 25, and 35 percent are typical (Andrews et al., 2012; Card et al., 2013; Alvarez et al., 2018). Manager and store effects are identified within connected sets of stores linked by moves; the largest covers about three-quarters of the panel, and every decomposition below normalizes within its set. The remaining concern is statistical rather than economic: fixed effects estimated from finite mover counts are noisy, and the noise inflates plug-in variance components. I address it with the leave-out estimator of Kline et al. (2020) (KSS).

Table 5: AKM variance decomposition with limited-mobility-bias corrections (log shrinkage value)

	<i>Baseline</i>	<i>Andrews</i>	<i>KSS</i>
$\widehat{\text{Var}}(\theta)$	0.049	0.009	0.016
$\widehat{\text{Var}}(\psi)$	0.287	0.243	0.251
$\widehat{\text{Cov}}(\theta, \psi)$	-0.014	0.002	-0.002
$\widehat{\text{Corr}}(\theta, \psi)$	-0.120	0.044	-0.038

Largest connected set of the general-manager panel. θ = manager fixed effect, ψ = store fixed effect; observation-weighted variances and covariances. *Baseline*: OLS variance of the estimated fixed effects. *Andrews*: homoskedastic correction of Andrews et al. (2008). *KSS*: leave-out estimator of Kline et al. (2020).

Table 5 reports the variance decomposition with and without the correction. In the plug-in decomposition, manager fixed effects account for 49.8 percent of the variance of log shrinkage value and store fixed effects for 42.4 percent—covariance shares, $\text{Cov}(y, \cdot)/\text{Var}(y)$, which allocate the sorting covariance and so need not mirror the raw variance components the table reports.⁸

The leave-out correction cuts the manager component to roughly a third of its plug-in value— $\widehat{\text{Var}}(\theta)$ falls from 0.049 to 0.016. The corrected component remains economically large: a manager one standard deviation better is associated with about 13 percent lower shrinkage ($\sqrt{0.016}$), holding the store and the month fixed. Appendix B plots the estimated fixed effects themselves; those plots convolve true effects with sampling noise, which is why the inference rests on the corrected components of Table 5 rather than on raw dispersion.

The classification is also insensitive to the estimation error that the correction quantifies. Managers are labeled by a median split within connected set, so the label depends only on ranks, and shrinking each fixed effect toward its cell mean in proportion to its noise reorders very few of them: an empirical-Bayes correction under the plug-in reliability reclassifies fewer than one percent of managers, and a correction under the reliability implied by Table 5 itself—roughly one third—reclassifies under two percent. Re-estimating every design that uses the label under the corrected classifications leaves the results unchanged: the arrival effect, the enforcement response, the presence response, and the complementarity differential retain 88 to 93 percent of their magnitudes at the same significance levels.

The decomposition assigns the manager a first-order role in shrinkage, of the same order as every permanent characteristic of the store combined. It is an accounting result, built on

⁸Every manager claim in the paper rests on the *KSS* column; the plug-in shares are quoted only to show what the correction changes.

fixed effects; the sharpest confirmation of the economic content of the manager component is in the next section, which leverages the firm's rotation of managers directly and asks whether shrinkage moves when the manager moves, in an arrival-and-departure design.

5 What Do Good Managers Do?

A central question that the paper answers is what good managers are doing across stores. This section exploits the high-frequency data on managers’ actions—the minute-level time clock, the personnel record, and the store’s operational panel—to ask what the managers whose fixed effect reduces shrinkage do differently inside the store. This section performs two validations of the fixed effects and then takes the arrival design across manager behavior.

Classifying managers I classify general managers into two types based on their AKM fixed effects for log shrinkage value, estimated in Section 4. A manager is high-type (HT) if her fixed effect falls below the median of the manager-effect distribution within her connected set; otherwise, she is low-type (LT).⁹

Design I use an arrival design throughout the section. An *arrival* occurs the first time a store comes under a manager of the given type for two consecutive months, and the first of those months is the event date; the two-month rule screens out temporary covers and end-of-month payroll overlaps. The period before the arrival is the reference period and we call it the handover month. A *departure* is the mirror image: the first month the store is no longer under that manager after the first arrival. We use a within-store treatment and test for robustness with staggered treatment designs. For a store-month outcome Y_{jt} I estimate

$$Y_{jt} = \sum_{\substack{k=-8 \\ k \neq -1}}^8 \beta_k \mathbf{1}[t - E_j = k] + \varphi_j + \lambda_{m(t)} + \xi_{y(t)} + \varepsilon_{jt}, \quad (21)$$

where E_j is store j ’s event date, φ_j is a store fixed effect, $\lambda_{m(t)}$ and $\xi_{y(t)}$ are calendar-month and year effects, and the endpoint dummies bin event times beyond ± 8 months. Standard errors are clustered at the store level. A companion difference-in-differences replaces the event-time dummies with a single $\text{Post}_{[0,8]}$ indicator for the first nine months after the event; that coefficient is the one reported in every table cell.

5.1 Validating the Classification of HT Managers

First, I validate the HT management classification; if the AKM fixed effect carries economic meaning, the arrival of an HT manager should reduce shrinkage. Using the event study design from equation 21, I find that shrinkage falls 6.8 percent over the nine months following a

⁹A lower fixed effect means less shrinkage under her management, so high-type managers are the better half of the distribution.

high-type manager’s arrival and rises 6.5 percent after her departure, reversing essentially the whole arrival reduction. The low-type designs mirror these results: shrinkage rises by 4.9 percent when a low-type manager arrives and shows no significant change when she departs. Appendix C reports these same four designs for shrinkage-to-sales, with the same pattern.

Table 6: Shrinkage on manager arrivals and departures

	Log shrinkage value
Arrival of a high-type manager	−0.0683*** (0.0146)
Departure of a high-type manager	0.0653* (0.0336)
Arrival of a low-type manager	0.0489*** (0.0176)
Departure of a low-type manager	−0.0343 (0.0305)
Mean	12.43
Median	12.35

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a difference-in-differences regression of the store-month outcome on a single post indicator equal to one in the first nine months after the event (event months 0 to 8) and zero in every other month, with store, calendar-month, and year fixed effects and standard errors clustered at the store, on the log value of inventory losses. Mean and Median are of the log outcome. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Under staggered timing, the pooled coefficient in Table 6 provides a conservative measure: stores treated earlier act as controls for stores treated later while their own shrinkage is still depressed, which attenuates the estimate (Goodman-Bacon, 2021). However, restricting the same regression to the event window yields a much stronger result: shrinkage falls by 19 percent after a high-type arrival, aligning closely with the depth of the event-study path.

Log shrinkage value

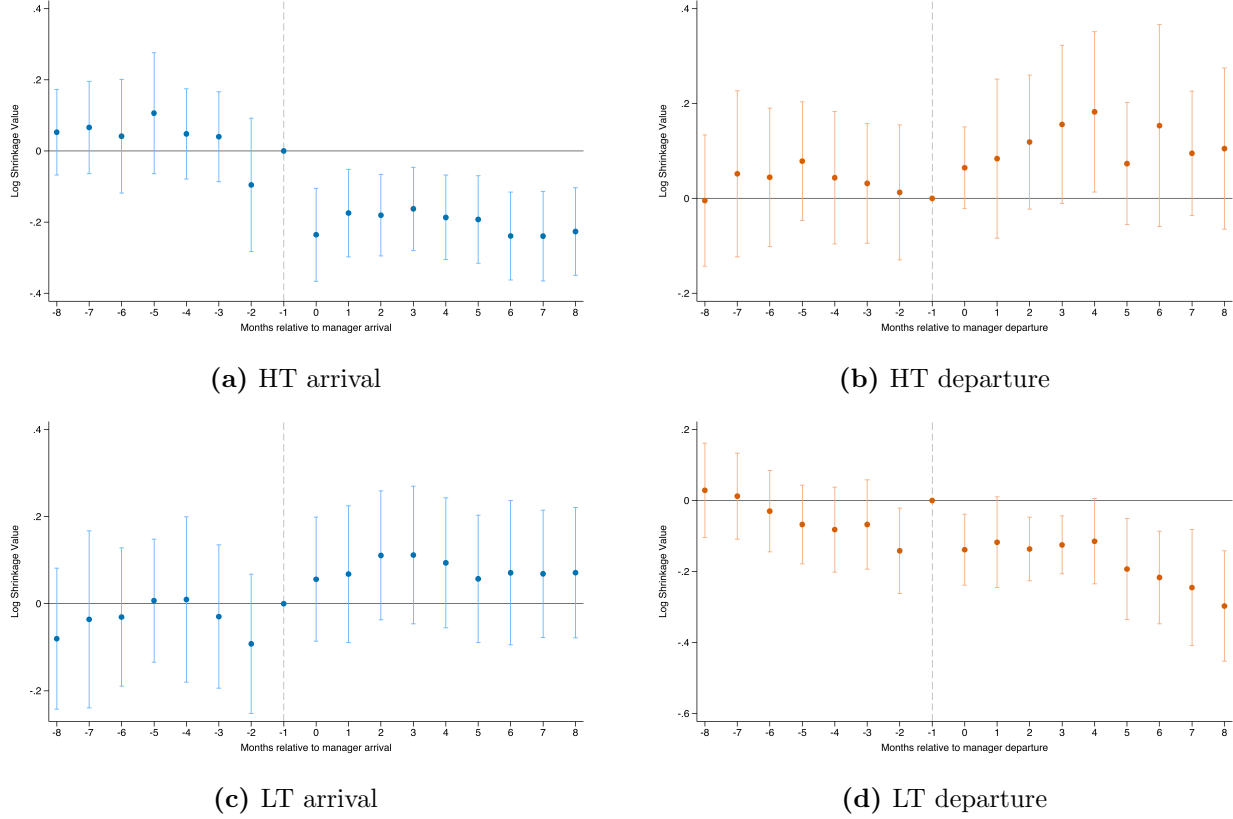


Figure 1: Log shrinkage value around the four manager transitions

Notes: Each panel plots the coefficients β_k from an event-study regression of the store-month outcome on event-month indicators, with store, calendar-month, and year fixed effects, for one of the four transition designs, estimated on that design's own event sample: (a) the arrival of a high-type manager, (b) the departure of a high-type manager, (c) the arrival of a low-type manager, and (d) the departure of a low-type manager. A manager is classified as high-type (HT) if her fixed effect in the decomposition of log shrinkage value into manager, store, and month fixed effects, lies below the median of the connected set—her stores lose less merchandise under her than under the median manager—and low-type (LT) otherwise. The outcome is the store's log shrinkage value in the month; the unit of observation is the store-month. The month before the event, the handover month ($k = -1$), is the omitted category. Store, calendar-month, and year fixed effects; ever-treated stores of each design. The vertical line marks the handover month. Bands are 95 percent confidence intervals with standard errors clustered by store.

Validating Management with Other Shirking Margins A manager's shrinkage fixed effect could reflect careful recordkeeping rather than management. Conversely, an HT manager in shrinkage can capture a broader quality across outcomes. To test this, I re-estimate the decomposition of Equation (19) separately for five additional outcomes—two alternative shrinkage definitions, turnover, and two non-justified-absence measures, the latter three recorded by payroll and the time clock, outside any manager's inventory paperwork—and compare each manager's fixed effects across outcomes. I find that the fixed effects from shrinkage are highly correlated with the fixed effects on other organizational characteristic

measures. In other words, HT managers on shrinkage are also high-type managers in other measures.

Figure 2 plots each manager’s fixed effect in every outcome against her fixed effect on every other one, and every pairwise correlation is positive: the three shrinkage-based effects are strongly correlated (0.63–0.85), and the shrinkage-value effect moves together with the manager effects in turnover (0.38) and in non-justified absence (0.65–0.82). High-type managers carry better fixed effects in every other outcome—strongly so for the shrinkage normalizations and turnover ($p < 0.001$), more weakly for the two absence measures ($p = 0.06$ – 0.09). If the fixed effect rewarded recordkeeping rather than conduct, it would not travel to margins the manager does not record. What the label captures is a broad dimension of managerial quality, in line with the manager-specific productivity effects of Lazear et al. (2015) and Bertrand and Schoar (2003), and this supports reading the arrival effects that follow as the effect of a better manager. The arrival effect is also graded rather than binary—it scales with how much the arriving manager improves on the incumbent one and vanishes for arrivals that are not upgrades—a dose response and novelty placebo reported in Appendix D.2.



Figure 2: Manager fixed effects across six outcomes

Notes: Each panel plots one point per manager, her fixed effect in one outcome (horizontal axis) against her fixed effect in another (vertical axis). Each fixed effect comes from a separate decomposition of that outcome into manager, store, and month fixed effects, estimated on the connected set, and is standardized to standard-deviation units. The six outcomes are log shrinkage value (first row and column), shrinkage per sales, shrinkage per worker, turnover, and two measures of non-justified absence, one from payroll and one from the time clock. Higher values mean more shrinkage, turnover, or absence. Pairwise Pearson correlations range from 0.38 (turnover) to 0.85 (across shrinkage definitions).

An additional check asks whether the classification is picking up something a personnel

file already shows—that the high-type managers are simply the senior ones, for instance, or the better paid. It is not. High- and low-type managers are statistically indistinguishable on the demographic characteristics that the firm records: gender (a difference of 2.4 percentage points, $p = 0.36$), schooling (0.16 years, $p = 0.39$), and age (0.98 years, $p = 0.07$). Tenure does not differentiate them either; if anything, high-type managers have been at the firm slightly less time, not more, and they are paid the same (Appendix D.6 reports every comparison). On paper, the good manager is ordinary; what the fixed effect captures is not a credential the firm already prices, which is precisely why identifying good managers from personnel records alone is hard.

5.2 Co-Presence and not a larger monitoring apparatus: Management as Monitoring

What does a good manager do differently once she arrives? The most direct way a manager can make lapses more likely to be caught is to watch the floor herself. She could instead put more watching into the store, by adding loss-prevention, supervisory, or security staff. I test both, and the answer is that she does the former and not the latter: after a high-type arrival the manager is present more and is on the floor with her workers more, while the store’s monitoring staff does not change.

The first part applies the arrival design to four measures of the manager’s presence, built at the store-month level from the minute-level time-clock records. *Manager shifts* counts the shifts the general manager works in the month; *weekend shifts* and *night shifts* count the subset worked on a Saturday or Sunday and at night. *Co-presence* is the share of frontline worker shift-hours worked while the general manager is on the clock, computed by overlapping each worker’s shift with the manager’s daily presence interval. After a high-type arrival, the manager works more shifts, more of them on weekends and at night, and she spends more time on the floor with her workers.

Table 7 reports the average effect over the nine months after each event, and Figure 3 shows its path month by month around the high-type arrival. The manager is there more. She works 1.5 more shifts per month on a base of 23.6, with weekend shifts up 0.39 on a base of 7.0, and night shifts up 0.66 on a base of 6.0, which translates to increases of five to eleven percent. Because she is present more while the store operates no differently, the frontline ends up being supervised more directly: co-presence rises 3.2 percentage points on a base near 57 percent. In the figure, all four measures rise in the month of the arrival and stay up through the eighth month, while the previous months scatter around zero. None of this happens when a low-type manager arrives: she moves none of the four measures. The

departure columns all shrink toward zero, consistent with the presence margin switching off when she goes. The added presence is broad rather than targeted at particular groups of workers (Appendix D.4).

Table 7: Managerial presence and manager-worker co-presence

	(1) Manager shifts/month	(2) Manager weekend shifts	(3) Manager night shifts	(4) Co-presence (worker-hrs)
HT manager arrival	1.5380*** (0.4739)	0.3923** (0.1583)	0.6565** (0.2741)	0.0324*** (0.0121)
HT manager departure	-0.3205 (1.6438)	-0.1342 (0.5279)	-0.5222 (0.6668)	-0.0238 (0.0385)
LT manager arrival	0.3250 (0.7416)	0.1401 (0.2325)	0.4689 (0.3773)	0.0039 (0.0160)
LT manager departure	0.2035 (1.0316)	-0.0832 (0.3287)	0.3389 (0.5016)	-0.0038 (0.0276)
Mean	23.58	6.99	6.04	0.575
Median	26.00	8.00	5.00	0.654

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a difference-in-differences regression of the store-month outcome on an indicator equal to one in the first nine months after the row’s event (event months 0 to 8) and zero in every other month, with store, calendar-month, and year fixed effects and standard errors clustered at the store, on the stores that experience the row’s event, estimated on store-month measures built from shift-level time-clock records. “Manager shifts” counts the general manager’s worked shifts in the month; “Manager weekend (night) shifts” the subset on Saturday–Sunday (night hours). “Co-presence” is the mean share of a worker shift’s hours overlapping the manager’s daily presence interval. Mean and Median are of the dependent variable. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

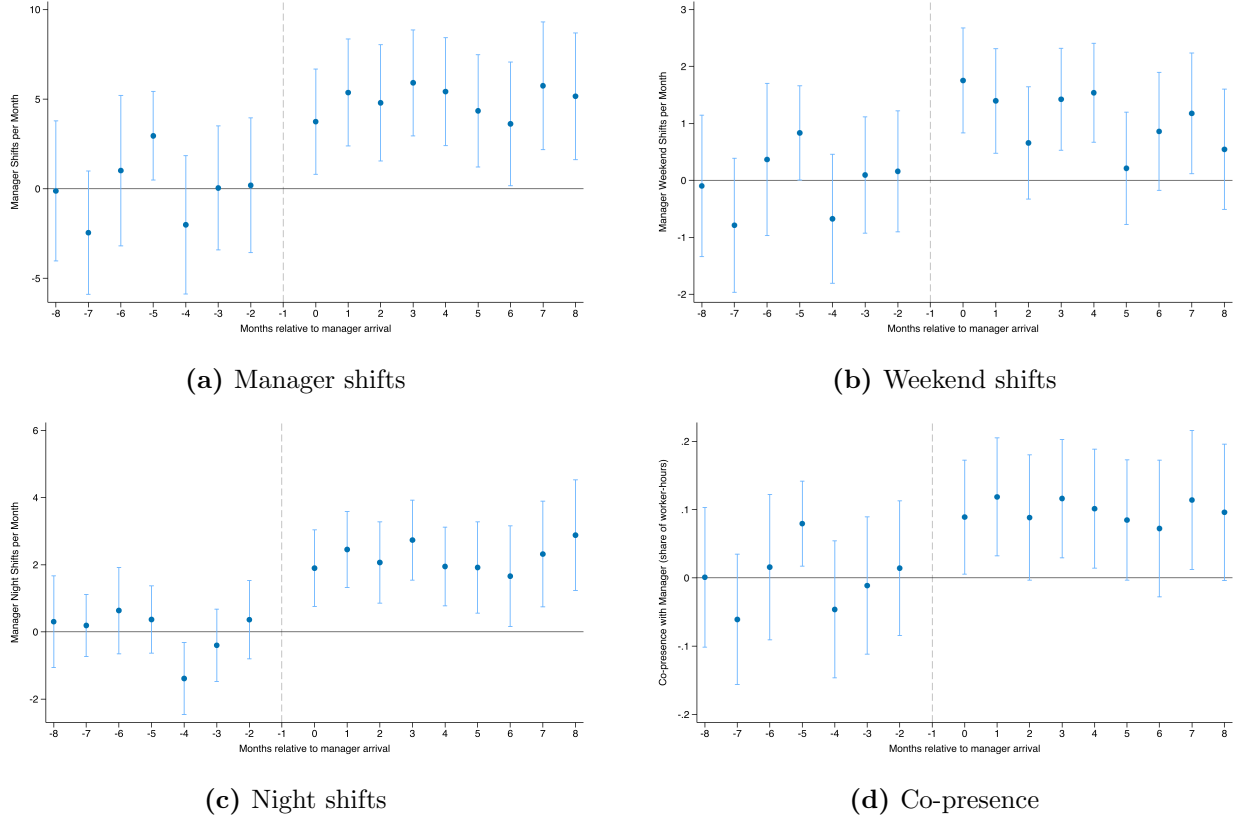


Figure 3: The manager’s presence around the high-type arrival

Notes: Each panel plots the coefficients from an event-study regression of the store-month outcome on event-month indicators, with store, calendar-month, and year fixed effects, around the arrival of a high-type manager for one measure of her own presence, all taken from the store’s biometric time clock: (a) the shifts she works in the month; (b) her weekend shifts; (c) her night shifts; and (d) co-presence, the share of frontline worker-hours worked while she is also on the clock. The unit of observation is the store-month. The omitted period is -1 . Store, calendar-month, and year fixed effects; ever-treated stores. The vertical line marks the handover month. Bands are 95 percent confidence intervals with standard errors clustered by store.

The manager then supplies the monitoring herself. The second part asks whether she also adds monitoring staff to the store. I apply the same arrival design to three measures of the store’s monitoring apparatus: *loss-prevention staff*, the workers in the store’s receiving, protection, and inventory-control positions per hundred workers; *supervisors*, area leaders, assistant managers, and managers as a share of the store’s headcount; and *security staff*, the share of the headcount in security positions. Table 8 reports the results. After a high-type arrival, loss-prevention staff move by 0.25 per hundred workers on a base of 5.2, the supervisory share by -0.25 on a base of 25.7, and the security share by 0.08 on a base of 13.1. None is distinguishable from zero, and the intervals rule out the kind of staffing change that could explain a shrinkage reduction of this size. A low-type arrival does not move them either, apart from a small, marginally significant fall in loss-prevention staff.

Measured in shifts rather than headcount, the three margins do not move after a high-type arrival (Appendix D.5).

Presence is the empirical content this paper gives the monitoring term μ_m of the model: the manager makes the dismissal threat credible by being there to notice, in line with the evidence that monitoring deters opportunism (Nagin et al., 2002) and that workers lift effort when observed (Mas and Moretti, 2009).

Table 8: The store’s monitoring apparatus on manager arrivals

	(1) Loss-prevention staff (per 100)	(2) Supervisors (% of staff)	(3) Security staff (% of staff)
HT manager arrival	0.2496 (0.1583)	−0.2454 (0.4051)	0.0750 (0.1834)
LT manager arrival	−0.2071* (0.1117)	0.0391 (0.4212)	0.1290 (0.1333)
Mean	5.19	25.69	13.15
Median	4.71	25.00	12.96

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a difference-in-differences regression on an indicator equal to one in the first nine months after the row’s event (event months 0 to 8) and zero in every other month, with the store-month outcome in the column: loss-prevention staff per hundred workers, and the shares of supervisory and security personnel in the store’s headcount. Store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores. Mean and Median are of the dependent variable.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3 The Team: Reorganized, Not Replaced

A good manager could also cut shrinkage by changing who works in the store—letting go of the workers who supply little care and replacing them with better ones. I test this with four measures of the team: turnover, the log of the month’s separations per employee; upward moves, within-store reassignments to a higher tier; cross-area moves, reassignments across functional areas, both per hundred workers; and team size, the log of the store’s headcount. The answer is that an HT manager reorganizes the team she inherits rather than replacing it. In her first months she moves existing workers up a tier and across areas. Separations rise only later—they are quits replaced one for one rather than dismissals, and headcount never changes. Because shrinkage falls in the arrival month, before turnover moves, the decrease does not come from dismissing workers.

Table 9 summarizes the results, while Figure 4 shows the dynamics. Turnover rises 17 percent over the nine months after a high-type arrival, yet team size does not move. The

rise is quits (+0.17) and one-for-one replacement hires (+0.21), with no detectable rise in dismissals (+0.003, standard error 0.079; Appendix Table 19 reports the full decomposition). What she does with the workers she keeps is modest and internal: reassignments to a higher tier rise by 0.19 per hundred workers on a base of 1.0 and reassignments across functional areas by 0.21 on a base of 1.4, while the composition of the team—its diversity of age, tenure, education, gender, tier, and pay—remains unchanged (Appendix D). The cross-area moves concentrate in reassignments into client-facing roles, with moves into back-office roles, positive but imprecise, and no rise in moves into the loss-prevention chain (Appendix D.3); she redeploys the workers she has rather than staffing up inventory control. A low-type arrival moves none of these margins. Appendix D traces the other three transitions: a low-type arrival produces the same fourth-month increase in separations but only a modest, imprecise rise in reassignments, and after a low-type departure upward and cross-area reassignments increase.

Table 9: The team on manager arrivals and departures

	(1) Log turnover (p.e.)	(2) Upward moves (per 100)	(3) Cross-area moves (per 100)	(4) Log team size
HT manager arrival	0.1738** (0.0724)	0.1920*** (0.0651)	0.2076** (0.0920)	0.0074 (0.0108)
HT manager departure	0.0519 (0.1896)	0.1260 (0.1608)	0.2021 (0.1993)	−0.0081 (0.0248)
LT manager arrival	0.0245 (0.0610)	−0.0048 (0.0702)	0.0286 (0.1134)	−0.0089 (0.0091)
LT manager departure	0.0279 (0.1486)	0.3125** (0.1423)	0.5081*** (0.1896)	0.0065 (0.0193)
Mean	−3.32	1.01	1.44	4.12
Median	−2.80	0.00	0.79	4.03

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a difference-in-differences regression of the store-month outcome on an indicator equal to one in the first nine months after the row's event (event months 0 to 8) and zero in every other month, with store, calendar-month, and year fixed effects and standard errors clustered at the store, on the stores that experience the row's event. Turnover is the log of the monthly separation rate per employee; upward moves are within-store reassignments to a higher tier while cross-area moves are reassignments across functional areas, both per hundred workers. Mean and Median are of the dependent variable. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

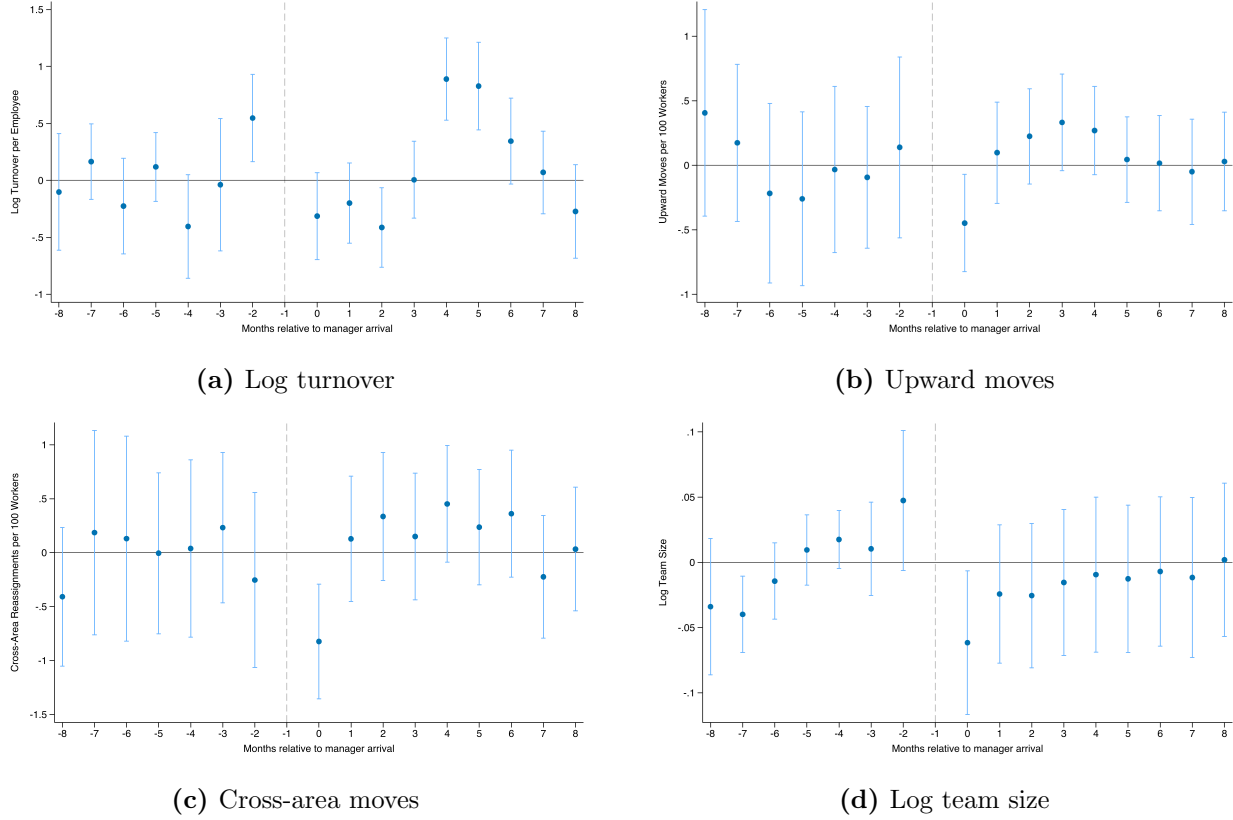


Figure 4: The team around the high-type arrival

Notes: Each panel plots the coefficients from an event-study regression of the store-month outcome on event-month indicators, with store, calendar-month, and year fixed effects, around the arrival of a high-type manager for one measure of the store's team, from the personnel movement log: (a) log turnover, the monthly separation rate per employee; (b) upward moves, reassignments to a higher tier within the store, per hundred workers; (c) cross-area moves, reassignments across functional areas, per hundred workers; and (d) log team size. The unit of observation is the store-month. The omitted period is -1 . Store, calendar-month, and year fixed effects; ever-treated stores. The vertical line marks the handover month. Bands are 95 percent confidence intervals with standard errors clustered by store.

6 Wage Premia and Shirking: Two-Way Fixed-Effects Evidence

The manager is one lever a firm holds over care; pay is another. A worker whose job pays more than she could earn elsewhere has something to lose if she is dismissed, and the threat of dismissal disciplines her care only to the extent of that rent. The model in Section 3 therefore predicts that shrinkage falls as the wage premium rises. This section tests that prediction.

Because the firm sets pay by position, the premium a worker earns moves only with the outside wage of her municipality. Outside wages vary widely across Mexico: the middle 90 percent of municipalities span about USD 150 to USD 440 a month in mean market wage (Appendix Figure 18), so the same position commands a very different premium depending on where the store is, and that premium moves over time as local wages move. I use this variation at two levels. At the store level, I compare stores that pay different premia in the same municipality and quarter; this is the level at which shrinkage is recorded, and the level of the paper’s central outcome. At the worker level, I follow the same worker as the outside wage of her cell moves. The worker level answers a question that the store level cannot. A store that pays a higher premium may simply employ different workers—a premium can attract and retain more careful people—so a store-level gradient could reflect who works in the store rather than how a worker responds to her rent. The model’s claim is about the second: the same worker supplies more care when she has more to lose. Following one worker, with worker and store fixed effects, removes every permanent attribute of the worker and of her store and isolates that response. Shrinkage is not recorded by worker, so this level is estimated on the complementary shirking margins—absence, turnover, shifts, and hours.

The store-level model is

$$\log Y_{it} = \delta \log W_{it}^p + Z_{it}\theta + \eta_{m(i)} + \tau_t + \nu_{it}, \quad (22)$$

where Y_{it} is inventory shrinkage of store i in quarter t ; W_{it}^p is the store wage premium, the average of the worker premium in Equation 1 across the store’s workers; $\eta_{m(i)}$ is a municipality fixed effect; τ_t is a year-by-quarter fixed effect; and Z_{it} optionally includes store format and log sales. Standard errors are clustered by municipality. The fixed effects absorb permanent differences across municipalities and shocks common to all stores in a quarter, so δ rests on movements in the local outside wage rather than on its level. Table 10 reports the estimates. The shrinkage gradient in column 1 is -0.113 : a 10 percent lower premium is associated with about 1.1 percent more shrinkage. The gradient keeps its sign and significance when

I add store format and log sales and when I replace the year-by-quarter fixed effects with separate year and quarter effects (columns 2–4), and it is robust to normalizing shrinkage by sales, in value and in physical units (Appendix Table 28).

Table 10: Store-level TWFE: inventory shrinkage (Log–Log)

<i>Wage premium (Log)</i>	(1)	(2)	(3)	(4)
Coefficient	−0.113***	−0.114***	−0.045**	−0.111***
SE	(0.032)	(0.033)	(0.021)	(0.031)
Store format	No	Yes	Yes	No
Log sales	No	No	Yes	No
Municipality FE	Yes	Yes	Yes	Yes
Year×Qtr FE	Yes	Yes	Yes	No
Year + Quarter FE	No	No	No	Yes
Mean	12.41	12.41	12.41	12.41
Median	12.35	12.35	12.35	12.35

Store-quarter level. The dependent variable is the log value of inventory losses in Mexican pesos; Mean and Median are of the log outcome. Standard errors clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The complementary shirking margins move the same way (Appendix E). At the store level, a 10 percent lower premium is associated with about 1.3 percent more non-justified absenteeism, 1.6 percent more turnover, and 0.4 percent more total absenteeism, and with 0.3 percent fewer monthly shifts and 0.1 percent fewer daily hours (Appendix Table 29). Within the same worker, a larger premium goes with fewer absence days, total and non-justified, with lower odds of any absence, of any non-justified absence, and of separation, and with more monthly shifts and longer daily hours; every estimate keeps its sign and significance under controls for managerial tier, tenure, and age and under the choice of time fixed effects (Appendix Tables 31–32). For the margins observed at both levels, the signs match, and the magnitudes are close for absence days, shifts, and hours, while the non-justified absence gradient is larger at the store level than within the worker (Appendix Table 33).

6.1 Could the Outside Wage Move Shrinkage Directly?

The premium moves because the outside wage moves, so the gradient is the effect of the rent only if the outside wage does not reach shrinkage by another road. I consider six.

The incentive to take merchandise from outside. Part of what goes missing from a store is taken by people who do not work there, and their incentive depends on what they can earn legally; where local wages rise, the return to legal work rises and property crime falls (Gould et al., 2002; Machin and Meghir, 2004). This channel works against the estimate.

Shrinkage is written as the sum of a part that falls with the premium and a part that falls with the local wage. A higher outside wage lowers the premium, which raises the first part, and lowers the second. The estimated gradient is the rent effect net of the outside-theft effect, so it understates the rent effect in absolute value.

Local crime. A local boom could still move crime for other reasons. It does not move with the premium: the premium does not predict commercial robbery, homicide, or the share of residents who report feeling unsafe in the store’s municipality, and the shrinkage gradient is unchanged when I control for them (-0.114 with the crime measures, -0.139 adding perceived insecurity, against -0.113 in the baseline).

Local unemployment. A slack local labor market lowers what a worker can expect outside and could change who shops and who works in the store. Controlling for the unemployment rate of the store’s local labor market leaves the gradient at -0.120 .

Local prices. Shrinkage is recorded in Mexican pesos, so a local wage increase that raises local prices would raise the value of the same missing merchandise. The gradient is not a price effect: measured in physical units it is larger, -0.176 , and it is -0.146 for units lost per unit sold.

Product mix. Where local incomes are higher a store may carry more perishable merchandise, which is lost more easily. The premium does not predict the perishable share of the store’s inventory.

Sales. A local boom raises sales, and a store that sells more has more to lose. The gradient survives the control: it is -0.069 conditional on log sales and -0.045 adding store format (Table 10, column 3), and -0.062 for shrinkage as a share of sales (Appendix Table 28).

7 Managers and the Wage Premium: Complements in Discipline

Care is attention a firm cannot measure and a supervisor observes only imperfectly. It cannot be traced to any particular worker. What the firm records is a loss traced to a unit; a store, a restaurant, a plant. The monitor is therefore indispensable; unlike absence, punctuality, or the length of a break, care leaves no record unless someone is there to observe it. The worker, in turn, cannot be paid for it directly; there is no pay for performance on an outcome that cannot be traced to her, so the premium the firm pays over the local market is the incentive it has left. Sections 5 and 6 show that each lever works: shrinkage falls when a better manager arrives, and it falls where the premium is higher. The question is whether each can sustain care independently or whether the two are more effective when deployed jointly.

The question matters because the two levers do not cost the same. In the average store, with about 67 workers and a premium of USD 93 a month each, raising the premium by 10 percent costs about USD 625 a month. The gradient of Section 6 associates that raise with 1.1 percent less shrinkage; about USD 190 a month. On the shrinkage margin alone, a dollar of premium returns about thirty cents. A high-type manager, in contrast, costs the same as a low-type one, because pay is fixed by position; the cost of better management is the cost of developing managers, which at least theoretically is lower than the cost of raising the wage of every worker in the store. If the two levers are substitutes, a firm can economize on the expensive one. If they are complements, it cannot; the manager needs a rent to threaten, and the rent needs a monitor.

This section estimates the arrival design of Equation (21) separately in high- and low-premium environments. The arrival of a high-type manager reduces shrinkage by 12.6 percent where the premium is high and by 4.3 percent where it is low; a difference of 8.4 percentage points. In the average store, that difference is about USD 1,400 of merchandise a month. To obtain the same additional reduction through pay alone, the gradient of Section 6 implies a premium about three-quarters higher; about USD 4,600 a month in the average store.

7.1 Design

I take the high-type arrival design of Section 5 and ask one question of it: is the shrinkage reduction a high-type manager delivers larger where the store's wage premium is high?

I classify each store's municipality, in the month its manager arrives, as a high-wage-premium (HWP) or low-wage-premium (LWP) environment: HWP if the premium the firm pays over the local ENOE market wage lies above that municipality's own median over the

sample, LWP otherwise. Fixing the environment at arrival makes it pre-determined with respect to everything that follows and constant within a store’s event window. I then re-estimate the arrival difference-in-differences of Equation (21), interacting the $\text{Post}_{[0,8]}$ indicator with this environment, separately for high- and low-type arrivals.

I include one alternative design. It pools the high- and low-type arrivals in a single regression and interacts the post-arrival indicator with the type of the arriving manager and with the wage environment,

$$\begin{aligned} Y_{jt} = & \beta_1 \text{Post}_{jt} + \beta_2 \text{Post}_{jt} \times \text{HT}_j + \beta_3 \text{Post}_{jt} \times \text{HWP}_j \\ & + \beta_4 \text{Post}_{jt} \times \text{HT}_j \times \text{HWP}_j + \varphi_j + \lambda_{m(t)} + \xi_{y(t)} + \varepsilon_{jt}, \end{aligned} \quad (23)$$

with the fixed effects and clustering of Equation (21). The coefficient β_4 is the triple difference: how much more the distance between a high-type and a low-type arrival widens where the premium is high. The effect in each of the four cells is a linear combination of the coefficients.

7.2 Good Managers Accomplish More Where the Premium Is High

The complementarity is the difference between the two environments in what a high-type arrival accomplishes, and Table 11 reports this in the upper panel. A high-type arrival reduces shrinkage by 12.6 percent where the premium is high and by 4.3 percent where it is low, so the same manager accomplishes 8.4 percentage points more, about three times as much, where the premium gives her dismissal threat something to confiscate; the difference is significant at five percent. The lower panel repeats the exercise for low-type arrivals. Shrinkage rises by 5.9 percent where the premium is low and by 3.7 percent where it is high; the second estimate, and the difference of 2.2 points between the two, are not distinguishable from zero. A weak monitor is bad for the store, whatever it pays: with little detection, the rent has nothing to threaten on the care margin, so a high premium does not replace the manager.

The result does not depend on how shrinkage is measured or on how the environment is defined. Appendix H repeats the exercise with log shrinkage over sales and log shrinkage per worker: the difference between environments is 6.1 and 10.1 percentage points, significant at ten and at one percent. Under three alternative definitions of the wage environment, the sign of the difference and the ordering of the two environments are preserved (Appendix Table 43 and Appendix Figure 21).

Table 11: Manager arrivals by wage environment at arrival

	Log shrinkage value
HT arrival, high-premium (HWP)	−0.126*** (0.032)
HT arrival, low-premium (LWP)	−0.043** (0.018)
Difference (HWP − LWP)	−0.084** (0.037)
LT arrival, high-premium (HWP)	0.037 (0.033)
LT arrival, low-premium (LWP)	0.059*** (0.021)
Difference (HWP − LWP)	−0.022 (0.039)
Mean	12.43
Median	12.35

Post_[0,8] = 1 for event times 0 through 8, interacted with the wage environment of the store’s municipality in the arrival month (HWP = premium above the municipality’s median; LWP otherwise). Each panel reports the arrival effect in each environment and then their difference (the complementarity); high-type arrivals above the rule, low-type below. Store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores only. A manager is high-type if her fixed effect for log shrinkage value lies below the median of the manager-effect distribution within her connected set, and low-type otherwise; an arrival is the first time a store comes under a manager of the given type for two consecutive months, and the first of those months is the event date; the outcome is the log value of inventory losses. Mean and Median are of the log outcome. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 5 traces the monthly path behind Table 11. Shrinkage falls in both environments when a high-type manager arrives, and from the second month after the arrival it falls by more where the premium is high. After a departure the two environments do not differ.

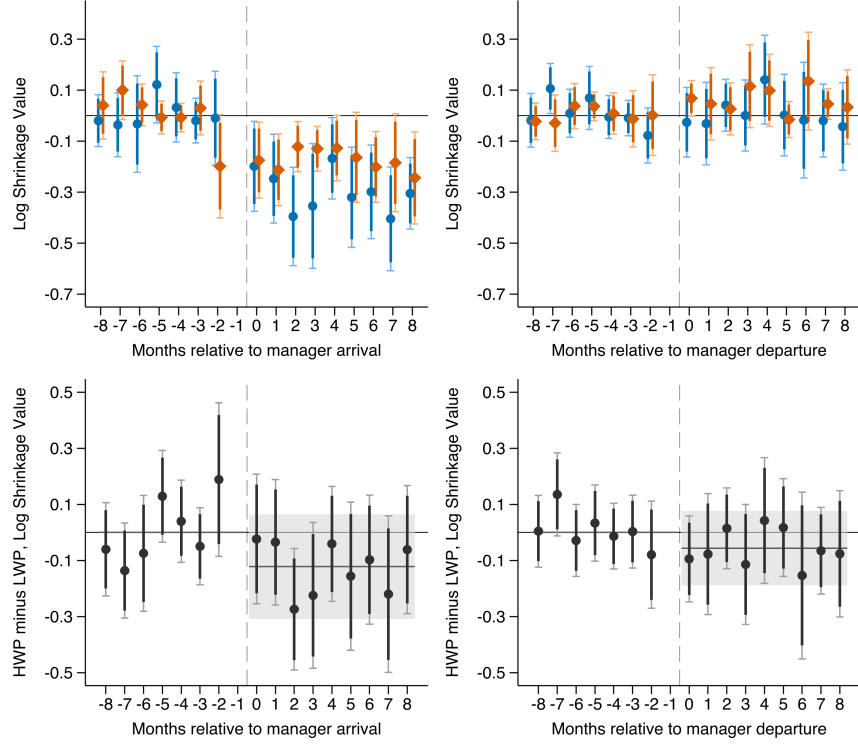


Figure 5: Log shrinkage value around high-type transitions, by wage environment

Notes: Event-study coefficients of log shrinkage value around the arrival (panel a) and the departure (panel b) of a high-type manager, estimated separately for high-premium stores (HWP, blue circles) and low-premium stores (LWP, orange diamonds); panels (c) and (d) plot the HWP minus LWP difference in each month. The wage environment is that of the store’s municipality two months before the event: HWP if its premium is above the municipality’s median, LWP otherwise. Coefficients are relative to the average of the eight months before the event; the coefficient of the handover month, the month before the event, is estimated but not plotted. Store, calendar-month, and year fixed effects. The vertical line separates the months before the event from the event month and the months after it. Thin whiskers are 95 percent confidence intervals and thick segments are 90 percent intervals, with standard errors clustered by store. In panels (c) and (d) the horizontal line and the shaded band are the average difference over the event month and the eight months after it, with its 95 percent interval.

The arrivals are staggered, so stores treated earlier serve as controls for stores treated later. Figure 6 re-estimates the arrival dynamics with the imputation estimator of Borusyak et al. (2024), which builds each treated store’s counterfactual from never-treated stores and from its own pre-arrival months only. The pattern is the same: shrinkage falls in both environments, and by more where the premium is high. The two paths coincide on impact; the high-premium path separates in the second and third months, where the two intervals do not overlap, and the two converge afterward. The complementarity also survives alternative ways of labeling managers. Rebuilding the label from the manager’s fixed effect on non-justified absence, an outcome the time clock records, gives a differential of -6.0 percentage

points, significant at five percent. Permuting residuals within store and re-running the entire pipeline one thousand times places the observed differential outside the full placebo distribution; net of the mechanical component it is -6.2 points. Estimating the label on odd months and testing it on even months gives -6.8 to -8.4 points (Appendices I and J).

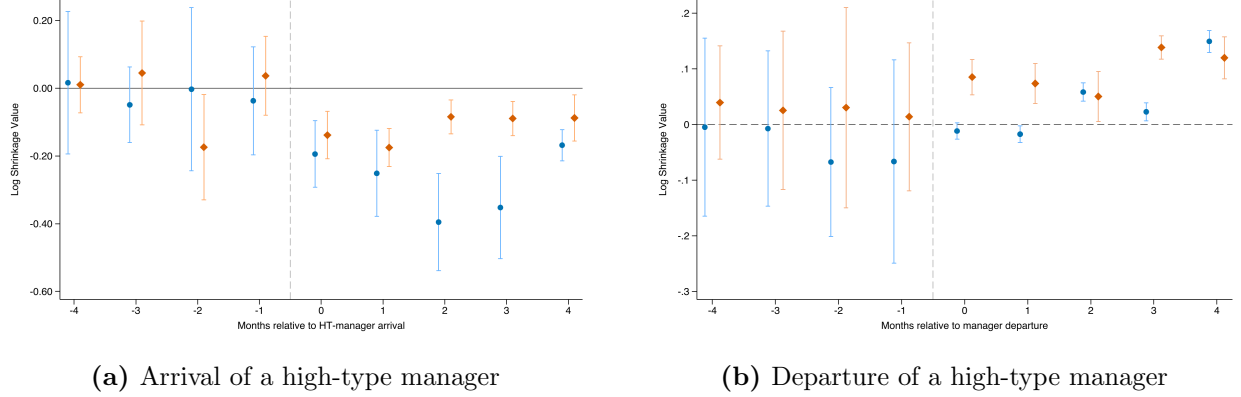


Figure 6: Log shrinkage value around high-type transitions, by wage environment: imputation estimator

Notes: Imputation estimator of Borusyak et al. (2024) for log shrinkage value around the arrival (a) and departure (b) of a high-type manager, estimated separately for high-premium (HWP, blue circles) and low-premium (LWP, orange diamonds) stores. The counterfactual for each treated store-month is built from never-treated stores and from the store’s own months before the event, so no already-treated store serves as a control and there is no omitted reference period: post-event points are deviations of observed from imputed shrinkage, and pre-event points are placebo coefficients estimated relative to months more than four months before the event. Only events whose pre-period is observed enter, far fewer than the events behind the pooled difference-in-differences estimates reported in the text. The unit of observation is the store-month. Whiskers are 95 percent confidence intervals with standard errors clustered by store; points are offset horizontally so the two series do not overlap.

Table 11 estimates high- and low-type arrivals in separate regressions, so each panel compares a store with itself and with the other stores of the same type. Figure 7 puts the four combinations of manager type and wage environment into the single regression of Equation (23). Estimated together, the four cells share the same controls and the same fixed effects, so they can be read against one another, and the gap between them is a coefficient the regression tests directly, the triple difference β_4 . The sample is also stricter: only arrivals whose months before the event are observed enter, with the stores without such an arrival as controls, so every cell compares a store before and after a manager it is seen to receive.

Figure 7 sets the firm’s two decisions side by side: whom it puts in charge of a store, and how much it pays its workers above the local market. A high-type manager lowers shrinkage wherever she arrives—by about 7.4 percent in a low-premium store and by about 20.4 percent in a high-premium one—so the same manager recovers almost three times as much where workers have more to lose. A low-type manager raises shrinkage wherever she

arrives—by about 10.1 percent where the premium is low and by about 5.3 percent where it is high—so a high premium does not turn the arrival of a weak monitor into a gain. All four effects are distinguishable from zero. The distance between the two types of manager is wider in the high-premium environment, the direction complementarity predicts, but high-premium arrivals are scarce and the figure does not measure that widening with precision; the powered test remains the per-type interaction of Table 11.

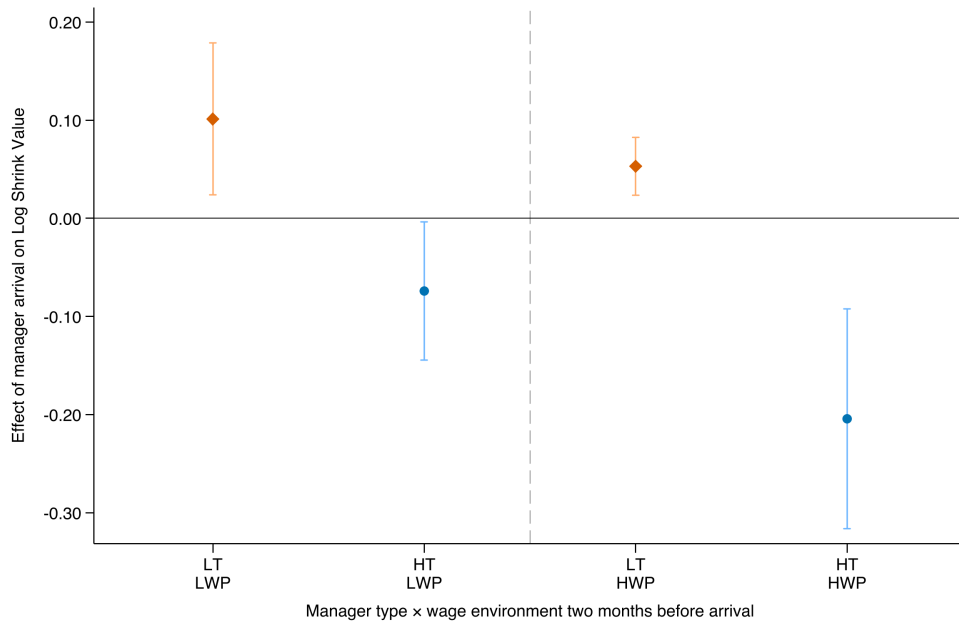


Figure 7: Effect of a manager arrival on log shrinkage value, by manager type and wage environment

Notes: Each point is the effect of a manager arrival on log shrinkage value in one of four cells, computed as a linear combination of the coefficients of a single pooled regression that interacts the post-arrival indicator with the type of the arriving manager and with the store’s wage environment: store, calendar-month, and year fixed effects. A manager is high-type (HT) if her fixed effect in the decomposition of log shrinkage value lies below the median of the connected set, so that her stores lose less merchandise under her than under the median manager, and low-type (LT) otherwise; the wage environment is high-premium (HWP) if the premium of the store’s municipality is above the municipality’s median and low-premium (LWP) otherwise. Whiskers are 95 percent confidence intervals with standard errors clustered by store; the dashed vertical line separates the two environments.

7.3 Does the Manager Do More Where the Premium Is High?

The complementarity admits two readings, which differ in what they imply for the manager’s conduct: a good manager could supply more discipline where the rent is high, or supply the same actions everywhere and have each action return more where the rent is high. To separate them, I re-estimate the arrival design with each action of Section 5 as the outcome, the presence and team margins as defined in that section, and ask whether the arrival changes

the action by more where the premium is high. For a store-month action a_{jt} the regression is the difference-in-differences companion of Equation (21), with the post-arrival indicator interacted with the wage environment at arrival as in Table 11,

$$a_{jt} = \beta \text{Post}_{jt} + \gamma \text{Post}_{jt} \times \text{HWP}_j + \varphi_j + \lambda_{m(t)} + \xi_{y(t)} + \varepsilon_{jt}, \quad (24)$$

where β is the arrival effect on the action in a low-premium store, $\beta + \gamma$ the effect in a high-premium store, and γ the differential between the two. Figure 8 plots γ for each action, divided by the standard deviation of the action so that margins measured in different units can be read on one axis; Table 12 reports the same estimates margin by margin, in the units of each action.

The pattern in the manager’s actions is a tendency rather than a finding. Her shifts per month, her weekend shifts, and her co-presence with the workers’ hours all rise by more in high-premium stores, but only the differential in shifts is significant, at ten percent, and the others are not; the data cannot say whether the manager is more present where the premium is high. Turnover shows the same tilt. It rises by 31 percent where the premium is high and by 12 percent where it is low, and the rise is in quits, which rise by 31 percent in high-premium stores and do not move detectably in low-premium ones, while dismissals do not move in either environment (Appendix Table 38). The differentials in turnover and in quits are not distinguishable from zero, so the comparison across environments is again a tendency. The one margin on which the two environments differ with precision is enforcement. Suspensions rise by 0.31 per hundred workers after a high-type arrival in a high-premium store and fall by 0.14 in a low-premium one; a differential of 0.45, significant at five percent. This is the mechanism of Proposition 4: the sanction is the manager’s own act and follows the rent, so where the premium gives her threat something to confiscate she uses it more. The team is not recomposed: team size, upward and cross-area moves, and the gender, occupational, and diversity mix of the store respond no differently across environments, with the supervisory share as the one exception among the manager’s inputs (Appendix G).

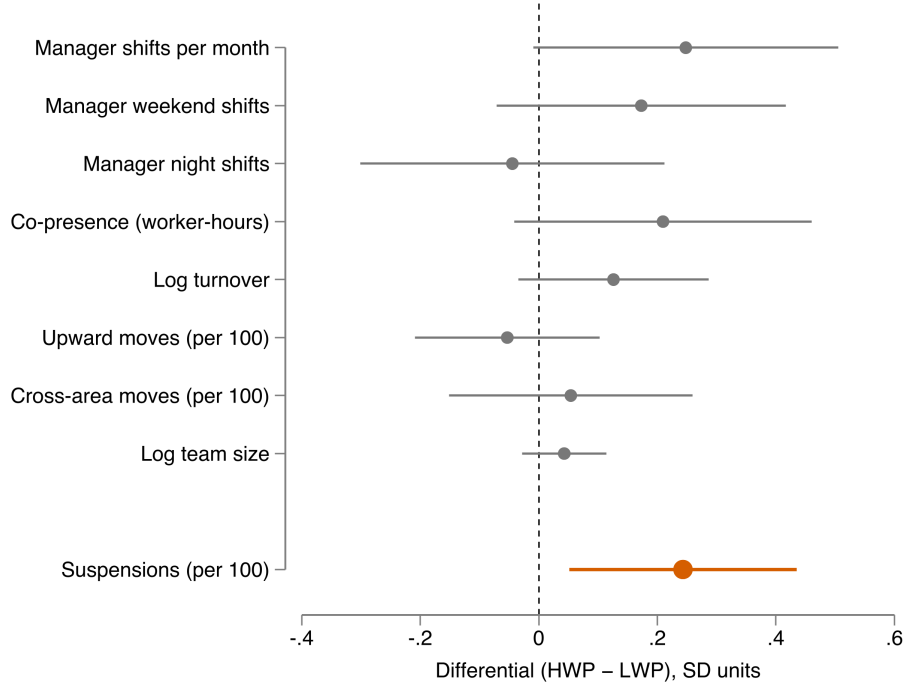


Figure 8: The differential between wage environments in each of the manager’s actions after a high-type arrival

Notes: Each point is the difference between high- and low-premium stores in the effect of a high-type manager’s arrival on one of her actions, the coefficient γ of Equation (24), divided by the standard deviation of the action so that the margins are comparable; a point to the right of zero means the arrival moves the action by more where the premium is high. A manager is high-type if her fixed effect in the decomposition of log shrinkage value lies below the median of her connected set, so her stores lose less merchandise under her than under the median manager; the wage environment is high-premium (HWP) if the premium of the store’s municipality in the arrival month is above the municipality’s median and low-premium (LWP) otherwise. Presence margins: shifts per month, weekend shifts, night shifts, and co-presence with frontline worker-hours. Team margins: log turnover, upward and cross-area moves per hundred workers, and log team size. Enforcement: suspensions per hundred workers, in orange. Store, calendar-month, and year fixed effects; standard errors clustered by store; ever-treated stores. Bars are 95 percent confidence intervals. The figure shows which actions differ between environments, not the size of the manager’s effect on any of them, which Table 12 reports.

Table 12: Does the manager do more where the premium is high? Every margin crossed with the wage environment at arrival

	Effect in LWP stores	Effect in HWP stores	Difference (HWP – LWP)
<i>Panel A. The manager's actions</i>			
Manager shifts per month	0.916	3.042***	2.126*
Manager weekend shifts	0.252	0.745***	0.493
Manager night shifts	0.715**	0.497	–0.218
Co-presence (worker-hrs)	0.020	0.063***	0.043
Log turnover (p.e.)	0.115	0.309***	0.194
Upward moves (per 100)	0.208**	0.131	–0.077
Cross-area moves (per 100)	0.173	0.272*	0.098
Log team size	0.001	0.023	0.023
<i>Panel B. Enforcement and the return</i>			
Suspensions (per 100)	–0.140	0.311**	0.451**
Log shrinkage value	–0.043**	–0.126***	–0.084**

Each row is a difference-in-differences regression of the indicated store-month outcome on a post-arrival indicator equal to one in the first nine months after the high-type arrival (event months 0 to 8) and zero in every other month, interacted with the wage environment fixed at the month of the high-type arrival (the municipality's premium above its own median): store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores. Columns: the arrival effect in low- and high-premium stores and their difference. Rates are per hundred workers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

What the subsection establishes is narrower than the two readings it opened with, and sharper. The manager may react to the rent with her presence: the tilt appears in every presence margin, but the differences are too imprecise to say that she does, and I do not claim this. What the data do say is that she enforces more where the premium is high, and that quits rise where she arrives at a high-premium store and not detectably where the premium is low. Both run through the rent. The suspension confiscates part of it, so it has bite only where there is something to confiscate; and a worker who will not supply the care demanded by a stricter monitor has, where the premium is high, a job worth keeping only by complying, so the arrival sorts some of them out. The complementarity in shrinkage is therefore the product described by the model, a monitor whose threat is worth more where the rent is higher, and not, as far as the data can tell, a manager who works harder where the firm pays more.

8 Minimum-Wage Hikes at the Border: Moving the Premium by Statute

Section 6 found that shrinkage falls where the wage premium is higher: a premium 10 percent lower is associated with about 1.1 percent more shrinkage. Section 7 found that the premium works through the manager, and accomplishes most where a good manager is in charge. Both results rest on movements in the local outside wage, which the firm does not control but which local conditions can move together with shrinkage. This section exploits the January minimum-wage hikes, which move the premium the firm pays over the legal floor by statute, and by more in the border zone than in the interior. The design adds to the evidence so far a shock whose size and date are set nationally, for reasons unrelated to any store. I then ask whether the manager on the floor changes the damage done by the hike. The model of Section 3 predicts that a small loss of rent moves care only where monitoring is weak: in a store whose manager already holds her workers at full compliance, the squeeze should leave shrinkage unchanged (Proposition 5).

Mexico's minimum-wage policy offers a clean way to move the wage premium by statute. The country keeps two statutory floors—a general one and a higher one in the northern border zone—and each January both rise by the same percentage, set by a federal commission and published in advance. The hikes move the premium in a way not chosen by the firm. Because the same percentage applies to a higher base, the step in Mexican pesos is larger at the border, so a single date moves border and interior stores by different amounts. Because the firm sets pay centrally by position and does not move its scale with the floor, the hike changes the premium its workers earn over the legal floor rather than a wage decided by the firm. And because the firm operates stores on both sides of the zone boundary, observed month by month, the same firm, under the same policies, is seen under two different shocks.

The exercise complements the two-way fixed-effects estimates of Section 6 in its source of variation and in the threat it removes. The gradient of Equation (22) is identified from movements in the ENOE outside wage of a store's municipality, net of municipality and year-by-quarter fixed effects. Its remaining threat is a local shock, a change in local demand or in local labor-market conditions, which moves outside wages and shrinkage together. The minimum-wage design moves a different component of the premium, the floor rather than the market wage, and moves it by statute; the size and the date of each hike are set nationally, for reasons unrelated to any store. The shock is therefore unrelated to local conditions by construction, and it is dated to a single month, so the months before it test whether border and interior stores were already diverging. What the design gives up is generality: it estimates the response to one kind of premium movement, in one part of the firm.

The exercise is an event study around each January hike. The step in Mexican pesos is about 1.5 times larger at the border and the firm’s scale is the same on both sides, so the border store’s premium over its floor compresses by more than the interior store’s. The border stores are the treated group, the interior stores of the same format are the control group, and the event is the January in which the new floor takes effect. I follow both groups from six months before to six months after each hike, and ask whether border shrinkage rises relative to interior shrinkage once the floor moves.

The border floor is recent. The government created the *Zona Libre de la Frontera Norte* in January 2019, and since 2020 the two floors have risen by the same percentage every January, which holds their ratio near 1.51. The hikes of January 2023 and January 2024 raised both floors by twenty percent, from permanently different bases (CONASAMI). About a quarter of the firm’s stores lie inside the border zone.

I estimate, for store k of format f in hike h at event month τ ,

$$Y_{kh\tau} = \alpha_{k,h} + \gamma_{f(k),m} + \sum_{s \neq -1} \beta_s (\text{Border}_k \times \mathbf{1}[\tau = s]) + \varepsilon_{kh\tau}, \quad (25)$$

where $Y_{kh\tau}$ is log shrinkage net of sales and the two hikes are stacked as separate experiments (Cengiz et al., 2019). The store-by-hike fixed effect $\alpha_{k,h}$ compares each store with itself within each hike, so no border–interior level difference enters the estimate. The format-by-calendar-month fixed effect $\gamma_{f(k),m}$ absorbs the seasonal pattern of each store format, so a border store is set against interior stores of its own format in the same month of the year. Border_k marks stores inside the border zone, $\mathbf{1}[\tau = s]$ marks event month s , and the month before the hike is the omitted category, so each β_s is the border–interior gap in month s relative to December. Standard errors are clustered by store. The summary replaces the event-month indicators with two, $\text{PostEarly}_{[0,2]}$ and $\text{PostLate}_{[3,6]}$, for months zero to two and three to six after the hike, each measured against the two months before it. The comparison follows the border-comparison tradition of the minimum-wage literature (Dube et al., 2010).

If the premium disciplines, a statute that compresses it should raise shrinkage. Figure 9 tests this. Panel (a) follows the border–interior difference in shrinkage month by month; panel (b) averages it over post-early, months zero to two after the hike, and post-late, months three to six. Post-early, border shrinkage is unchanged relative to the interior (-0.009 , $p = 0.60$). Post-late, it is 4.4 percent higher ($p = 0.022$). The timing is informative. The response is dated to the months in which the smaller premium persists, not to the day the law changes, as expected if the effect operates through the rent; the worker’s incentive to exert care falls as the value of keeping the job falls. Section 8.1 splits this response by the manager on the

same clock.¹⁰

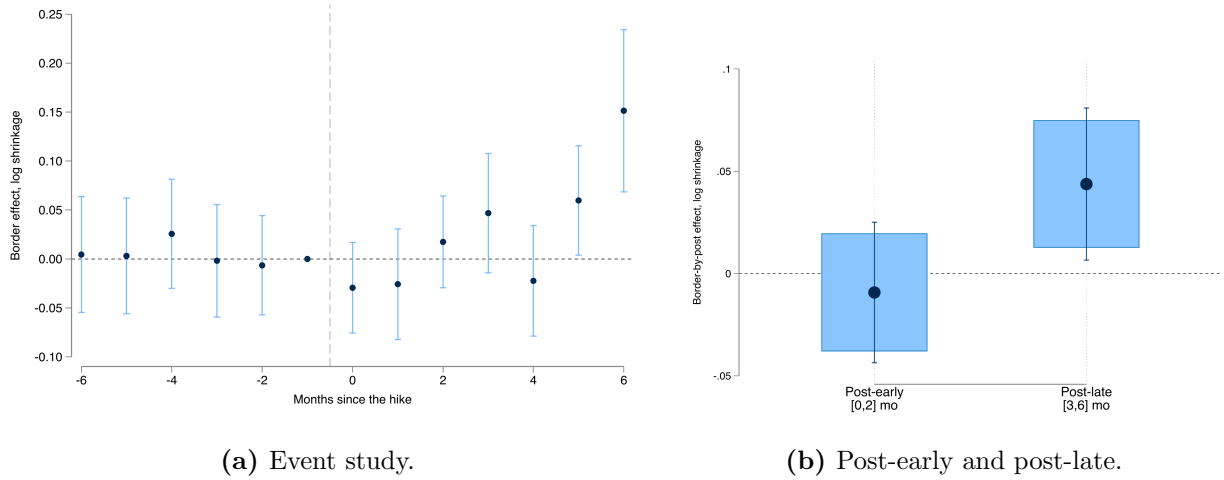


Figure 9: Shrinkage in border stores around the minimum-wage hikes

Notes: Panel (a) plots the border $\times$ event-month coefficients of a stacked event study of the two January hikes, with store-by-hike and format-by-calendar-month fixed effects, on log shrinkage net of sales: each point is the difference between border and interior stores in that month relative to the month prior to the January hike, which is the omitted category; the two January hikes in the sample are stacked as separate experiments and the vertical line marks the hike. Panel (b) plots the same difference averaged over post-early, months 0 to 2 after the hike, and post-late, months 3 to 6, each measured against the two months before the hike. Border stores are those in the zone with the higher statutory minimum. The unit of observation is the store-month; store-by-hike and format-by-calendar-month fixed effects; standard errors clustered by store. Whiskers are 95 percent confidence intervals; in panel (b) the box is a 90 percent interval.

The natural next test is the first stage: did the hike lower the wage premium as the paper measures it, the firm wage over the ENOE outside wage? That event study is underpowered. The outside wage is observed by quarter and the panel covers two and a half years, so the six months on each side of a hike contain two quarters, too few to trace the premium around the event. I instead show the part of the first stage that the firm's own records measure month by month: the worker's premium over her statutory floor. Figure 10 is the event study of Equation (25) at the worker level for the first of the two hikes, with that premium, in Mexican pesos per day, as the outcome, a worker fixed effect in place of the store effect, and a position-by-format-by-month fixed effect, so a border worker is compared with an interior worker in the same position and month. Both terms of the outcome are known exactly, the contractual wage from payroll and the floor from the statute, so it moves only when the floor or the firm's scale moves. The dashed line is the full statutory squeeze, the fall that would occur if the firm's wages did not move at all. The border worker's premium over her floor

¹⁰Post-late, the monthly path is uneven and climbs toward the end of the window, so the average over the four months is what should be read, not any single month; the design dates the response to the post-late months as a whole, not to a particular month.

drops in the month the new floor takes effect and stays down for the rest of the window, and the drop is about half the statutory squeeze. The premium therefore fell and remained lower, but by less than the statute implies: the 4.4 percent of Figure 9 is a response to the premium workers actually experienced, half the size of the one the statute implies.

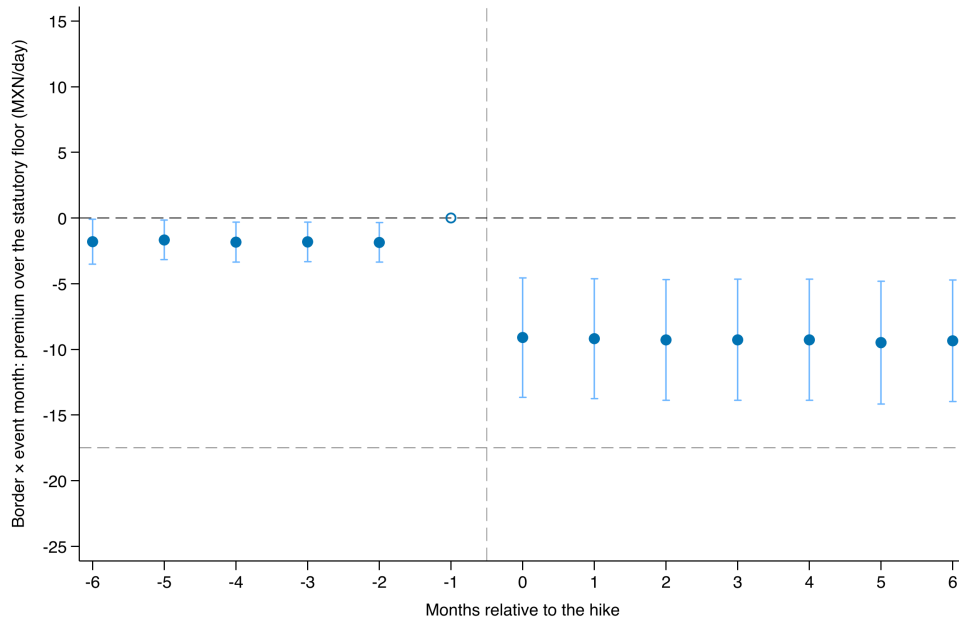


Figure 10: The worker’s premium over the statutory floor around the first January hike

Notes: Border $\times$ event-month coefficients on the worker’s contractual daily wage minus the statutory minimum of her zone, in pesos per day, around the first of the two January hikes. The sample is non-managerial full-time workers present throughout the window; the unit of observation is the worker-month. The month before the hike is the omitted category and is drawn as a hollow marker. Worker fixed effects and position-by-format-by-calendar-month fixed effects; standard errors clustered by store; whiskers are 95 percent confidence intervals. The dashed horizontal line is the full statutory squeeze: the difference between the border and interior floors’ steps, in pesos per day.

The question of whether the damage depends on the manager on the floor when the shock lands is taken up in Section 8.1.

8.1 The Monitor at the Moment of the Shock

This subsection brings the manager into the border design. Section 7 showed that a high-type manager reduces shrinkage by more where the premium is high, and Figure 9 showed that the hikes raise border shrinkage in the months after the hike. The question here is which border stores bear that rise: every store, or the stores where the manager is seldom on the floor? The level result does not settle it. In levels, a monitor’s threat is worth more where the rent is larger. A small cut in the rent is a different object. Where the manager is on the

floor and her workers already supply full care, a slightly smaller premium can still leave the job worth keeping by complying, so care need not move; where she is seldom present, care sits below that ceiling and responds to any change in what the job is worth. The model of Section 3 allows either pattern, so which one holds is an empirical question.

The design is the stacked border event study of Equation (25), summarized by the post-early and post-late indicators. I split the border stores into two groups by the general manager’s floor presence, measured from the biometric time clock in the December before each hike and divided at the median within format-by-hike cells, and interact each post indicator with the two groups. Both groups are compared with the same interior stores, so each group’s estimate is its own border–interior difference, and the object of interest is the difference between the two post-late responses. Because presence is measured before the hike, the hike cannot move the split. The cost is precision: each group holds half of the border stores, so the difference between the groups is estimated less precisely than the pooled effect of Figure 9.

Figure 11 shows the results. The two groups differ in both periods. Where the manager is on the floor, shrinkage falls post-early, by 4.9 percent relative to sales ($p = 0.10$), and the post-late months show no detectable effect (+1.9 percent); the erosion of the premium is, in effect, insured by the monitor. Where she is not, there is no post-early effect, and post-late shrinkage rises by 4.9 percent ($p = 0.02$). The level complementarity of Section 7 and this pattern are two readings of the same deterrent: a present manager makes the rent worth more, and keeps her workers far enough above the point of indifference that a small loss of rent does not move them.

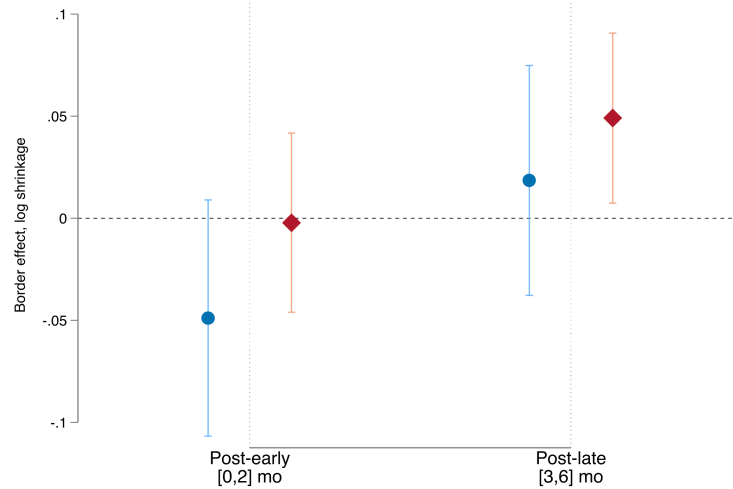


Figure 11: Shrinkage in border stores around the hikes, by the manager’s floor presence

Notes: Border $\times$ phase coefficients on log shrinkage net of sales, both January hikes stacked, from a stacked border-versus-interior regression with store-by-hike and format-by-calendar-month fixed effects, the post-hike months grouped into post-early and post-late periods, for two groups of border stores: those whose general manager is on the floor often (blue circles) and those whose manager is seldom there (red diamonds). Presence is measured from the store’s biometric time clock; the split is at the median presence within format-by-hike cells, on the manager in place the December before each hike. The horizontal axis has two periods, both measured against the two months before the hike: post-early, months 0 to 2 after the hike, and post-late, months 3 to 6. Store-by-hike and format-by-calendar-month fixed effects; whiskers are 95 percent confidence intervals, standard errors clustered by store.

9 Conclusion

In many jobs, effort takes the form of care: attention a firm cannot measure and a supervisor observes only imperfectly, since a lapse is caught only when someone is watching. The barista who gets an order right, the cook who keeps portions to the recipe, the line worker who lets no defective unit through, the clerk who keeps merchandise from going missing—each supplies care. What the firm sees is the residue: an order remade, food thrown out, a part scrapped, a gap on the inventory count; a cost it can record, but not trace to any particular worker. That is the firm’s problem with care. Because the loss cannot be traced to a particular worker, measures to control it are difficult to implement; a firm cannot do pay for performance for care it does not observe, nor charge the loss to whoever caused it. Since the residue is scattered across many small failures, it is rarely recorded systematically at all; where it is, it is large. In United States retail, inventory shrinkage—merchandise that goes missing from stores—reached \$112 billion in 2022, about 1.6 percent of sales, on the order of grocers’ net profit margin (National Retail Federation, 2023; Damodaran, 2026); the most recent global survey puts it at a similar 1.8 percent of sales worldwide (Tyco Retail

Solutions and PlanetRetail RNG, 2018). Retailers treat it as a security problem, where they lock merchandise behind glass, expand video surveillance, and increase loss-prevention budgets, while some chains have closed stores because of it (Selyukh, 2024). Despite the cost of lapses in care for firms, it remains unclear which levers a firm holds to sustain care, and there is no systematic way to analyze the problem that could guide a firm in choosing among them.

I study inventory shrinkage as a measure of care and analyze two of the levers a firm holds to sustain it: monitoring—through its managers—and the wage premium. Using manager rotations that are unrelated to store shrinkage, I estimate each manager’s effect on shrinkage and classify managers as high-type or low-type by their fixed effect within their connected set. Two validations support the classification. Shrinkage falls by close to 7 percent over the nine months after a high-type manager arrives at a store, and rises by about as much after she leaves (Figure 1). And the managers who reduce shrinkage also carry better fixed effects on margins they do not record; their effects on turnover and on non-justified absence, which payroll and the time clock record, comove with their effect on shrinkage, with correlations between 0.38 and 0.82 (Figure 2). These managers are not better bookkeepers; they appear to manage differently. When a high-type manager arrives, she is in the store more: she works about 1.5 more shifts a month, including more weekend and night shifts, and her co-presence with the frontline rises by about 3 percentage points (Figure 3). She reworks the team she inherits, moving workers up a tier and across areas, while turnover rises through quits rather than dismissals (Figure 4). She does not add security or supervisory staff, and on demographic characteristics high-type managers are indistinguishable from the rest.

I then use the firm’s policy of setting wages centrally by position, which makes a worker’s premium move with the outside wage of her municipality, to relate shrinkage to the wage premium. Shrinkage is lower where the premium is higher: a 10 percent lower premium is associated with about 1.1 percent more shrinkage. Because the two levers do not cost the same, whether they act as complements or substitutes matters for how a firm should use them; if they are substitutes, a firm can lean on the cheaper one, while if they are complements, the manager needs a rent to threaten and the rent needs a monitor. The evidence points to complements. After a high-type arrival, shrinkage falls by close to 12 percent where the premium is high against roughly 4 percent where it is low; a gap of eight percentage points. Where the premium is high, high-type managers also tend to be more present, although those differences are imprecisely estimated, and they enforce more: suspensions rise by 0.31 per hundred workers after a high-type arrival in a high-premium store and fall by 0.14 in a low-premium one, which is a significant difference at five percent.

Finally, I use Mexico’s January minimum-wage hikes. Both statutory floors rise by the

same percentage, so the step in Mexican pesos is larger at the border, and the firm's pay scale does not move with the floor, so the premium over the legal floor narrows more in border stores. Comparing border with interior stores of the same format around each hike, border shrinkage is unchanged in the first months and about 4 percent higher in the months that follow. The rise appears in stores whose manager is seldom on the floor and not in those where she is often present, a pattern the design shows but does not estimate precisely.

I read these facts through a shirking model of care (Shapiro and Stiglitz, 1984). The firm pays a fixed wage by position, so the only formal incentive is dismissal: a worker caught neglecting her job loses the premium she earns over her outside option. Whether she is caught depends on the manager's presence on the floor. The deterrent is therefore a product, the probability that a lapse is detected times the rent lost upon dismissal, and the product delivers the three facts. A better monitor disciplines care; a larger premium disciplines it too; and, under conditions the model states, the two are complements, so a given manager accomplishes more where the premium is high. Because the sanction is the manager's own act, the model also predicts that she sanctions where the job is worth keeping, which is where the suspensions rise.

For firms, the two levers are one decision: how much to pay above the market and whom to put in charge of a store are best chosen together. On the shrinkage margin alone, the premium is an expensive lever. In the average store, with about 67 workers and a premium of about USD 93 a month each, a 10 percent raise costs about USD 625 a month and is associated with about USD 190 less merchandise lost, about thirty cents per dollar. A high-type manager, in contrast, costs the firm what a low-type one costs, because pay is set by position; the cost of better management is the cost of identifying and developing managers, which at least in theory is lower than the cost of raising the wage of every worker in the store. And the same manager is worth more where the premium is high. The gap of 8.4 percentage points between high- and low-premium stores amounts to about USD 1,400 of merchandise a month in the average store; obtaining the same additional reduction through pay alone would take, by the premium gradient, a premium about three-quarters higher, equivalent to about USD 4,600 a month. Identifying, developing, and placing good managers is therefore a way to make an existing pay scale go further, and a pay scale set without regard to who manages each store leaves part of its return unrealized.

For policy, the minimum-wage findings suggest that a hike changes more than the wage bill. By narrowing the premium over the legal floor, it weakens one of the incentives a firm uses to sustain care, and the cost appears months later, on a margin that standard datasets do not record. How large that cost is depends on how the firm is organized; in this firm the rise in shrinkage appeared where the manager was seldom on the floor. Firms facing

a hike may therefore need to adjust how they manage, not only how much they pay, and evaluations that look only at employment and wages would miss that adjustment.

The paper contributes first to the evidence on how pay and monitoring interact. Coviello et al. (2022) establish a wage-by-monitoring complementarity: a minimum-wage increase raises the productivity of bound workers, and by more where monitoring is intensive. This paper finds the same complementarity from the opposite direction, moving the monitor rather than the wage: the same manager accomplishes more where the rent is large. It also measures the monitor differently, as an individual manager whose arrivals and presence on the floor are observed, rather than as a staffing ratio. Second, it adds an outcome: shrinkage, treated in economics as an accounting line, is read here as a measure of care and linked—store by store and manager by manager—to the personnel of the store. Third, it adds variation to the evidence that pay premia sustain discipline within firms (Cappelli and Chauvin, 1991; Emanuel and Harrington, 2024): the premium moves both with local labor markets and by statute, against the same audited outcome. Finally, it adds to the literature that recovers manager effects from rotation and asks what good managers do (Bertrand and Schoar, 2003; Lazear et al., 2015; Minni, 2026; Adhvaryu et al., 2026), by estimating the manager’s effect on a margin a supervisor observes only imperfectly and showing that it depends on the wage environment in which the manager operates. The same reading is available wherever a firm audits a loss it cannot trace to anyone: orders remade, food thrown out, parts scrapped.

References

- Abowd, John M., Francis Kramarz, and David N. Margolis**, “High Wage Workers and High Wage Firms,” *Econometrica*, 1999, 67 (2), 251–333.
- Adhvaryu, Achyuta, Parker Howell, Andrea Neyra-Nazarrett, Anant Nyshadham, and Jorge Tamayo**, “What Do Good Managers Do and What Do They Leave Behind?,” 2026. Working paper, Harvard Business School.
- Akerlof, George A.**, “Labor Contracts as Partial Gift Exchange,” *The Quarterly Journal of Economics*, 1982, 97 (4), 543–569.
- Alvarez, Jorge, Felipe Benguria, Niklas Engbom, and Christian Moser**, “Firms and the Decline in Earnings Inequality in Brazil,” *American Economic Journal: Macroeconomics*, 2018, 10 (1), 149–189.
- Andrews, Martyn J., Leonard Gill, Thorsten Schank, and Richard Upward**, “High Wage Workers and Low Wage Firms: Negative Assortative Matching or Limited Mobility Bias?,” *Journal of the Royal Statistical Society: Series A*, 2008, 171 (3), 673–697.
- , ———, ———, and ———, “High Wage Workers Match with High Wage Firms: Clear Evidence of the Effects of Limited Mobility Bias,” *Economics Letters*, 2012, 117 (3), 824–827.
- Autor, David H., Alan Manning, and Christopher L. Smith**, “The Contribution of the Minimum Wage to US Wage Inequality over Three Decades: A Reassessment,” *American Economic Journal: Applied Economics*, 2016, 8 (1), 58–99.
- Bandiera, Oriana, Iwan Barankay, and Imran Rasul**, “Incentives for Managers and Inequality Among Workers: Evidence from a Firm-Level Experiment,” *Quarterly Journal of Economics*, 2007, 122 (2), 729–773.
- Bertrand, Marianne and Antoinette Schoar**, “Managing with Style: The Effect of Managers on Firm Policies,” *Quarterly Journal of Economics*, 2003, 118 (4), 1169–1208.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess**, “Revisiting Event-Study Designs: Robust and Efficient Estimation,” *Review of Economic Studies*, 2024, 91 (6), 3253–3285.
- Breza, Emily, Supreet Kaur, and Yogita Shamdasani**, “The Morale Effects of Pay Inequality,” *Quarterly Journal of Economics*, 2018, 133 (2), 611–663.

- Cappelli, Peter and Keith Chauvin**, “An Interplant Test of the Efficiency Wage Hypothesis,” *Quarterly Journal of Economics*, 1991, 106 (3), 769–787.
- Card, David, Alexandre Mas, Enrico Moretti, and Emmanuel Saez**, “Inequality at Work: The Effect of Peer Salaries on Job Satisfaction,” *American Economic Review*, 2012, 102 (6), 2981–3003.
- , **Jörg Heining, and Patrick Kline**, “Workplace Heterogeneity and the Rise of West German Wage Inequality,” *Quarterly Journal of Economics*, 2013, 128 (3), 967–1015.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer**, “The Effect of Minimum Wages on Low-Wage Jobs,” *Quarterly Journal of Economics*, 2019, 134 (3), 1405–1454.
- Chetty, Raj, John N. Friedman, and Jonah E. Rockoff**, “Measuring the Impacts of Teachers I: Evaluating Bias in Teacher Value-Added Estimates,” *American Economic Review*, 2014, 104 (9), 2593–2632.
- Coviello, Decio, Erika Deserranno, and Nicola Persico**, “Minimum Wage and Individual Worker Productivity: Evidence from a Large US Retailer,” *Journal of Political Economy*, 2022, 130 (9), 2315–2360.
- Damodaran, Aswath**, “Operating and Net Margins by Industry Sector (United States),” NYU Stern School of Business, data as of January 2026 2026. Net margin, Retail (Grocery and Food): 1.32 percent.
- Dube, Arindrajit, T. William Lester, and Michael Reich**, “Minimum Wage Effects Across State Borders: Estimates Using Contiguous Counties,” *Review of Economics and Statistics*, 2010, 92 (4), 945–964.
- Dustmann, Christian, Attila Lindner, Uta Schönberg, Matthias Umkehrer, and Philipp vom Berge**, “Reallocation Effects of the Minimum Wage,” *Quarterly Journal of Economics*, 2022, 137 (1), 267–328.
- Emanuel, Natalia and Emma Harrington**, “The Payoffs of Higher Pay: Labor Supply and Productivity Responses to a Voluntary Firm Minimum Wage,” Staff Report, Federal Reserve Bank of New York 2024.
- Goldsmith-Pinkham, Paul, Peter Hull, and Michal Kolesár**, “Leniency Designs: An Operator’s Manual,” NBER Working Paper 34473, National Bureau of Economic Research 2025.

- Goodman-Bacon, Andrew**, “Difference-in-Differences with Variation in Treatment Timing,” *Journal of Econometrics*, 2021, *225* (2), 254–277.
- Gould, Eric D., Bruce A. Weinberg, and David B. Mustard**, “Crime Rates and Local Labor Market Opportunities in the United States: 1979–1997,” *Review of Economics and Statistics*, 2002, *84* (1), 45–61.
- Harrison, David A. and Katherine J. Klein**, “What’s the Difference? Diversity Constructs as Separation, Variety, or Disparity in Organizations,” *Academy of Management Review*, 2007, *32* (4), 1199–1228.
- Hoffman, Mitchell and Steven Tadelis**, “People Management Skills, Employee Attrition, and Manager Rewards: An Empirical Analysis,” *Journal of Political Economy*, 2021, *129* (1), 243–285.
- Kline, Patrick, Raffaele Saggio, and Mikkel Sølvsten**, “Leave-Out Estimation of Variance Components,” *Econometrica*, 2020, *88* (5), 1859–1898.
- Lazear, Edward P., Kathryn L. Shaw, and Christopher T. Stanton**, “The Value of Bosses,” *Journal of Labor Economics*, 2015, *33* (4), 823–861.
- Machin, Stephen and Costas Meghir**, “Crime and Economic Incentives,” *Journal of Human Resources*, 2004, *39* (4), 958–979.
- Marx, Benjamin, Vincent Pons, and Tavneet Suri**, “Diversity and Team Performance in a Kenyan Organization,” *Journal of Public Economics*, 2021, *197*, 104332.
- Mas, Alexandre and Enrico Moretti**, “Peers at Work,” *American Economic Review*, 2009, *99* (1), 112–145.
- Minni, Virginia**, “Making the Invisible Hand Visible: Managers and the Allocation of Workers to Jobs,” *Quarterly Journal of Economics*, 2026, *141* (3), 1871–1920.
- Nagin, Daniel S., James B. Rebitzer, Seth Sanders, and Lowell J. Taylor**, “Monitoring, Motivation, and Management: The Determinants of Opportunistic Behavior in a Field Experiment,” *American Economic Review*, 2002, *92* (4), 850–873.
- National Retail Federation**, “Shrink Accounted for Over \$112 Billion in Industry Losses in 2022,” National Retail Federation, National Retail Security Survey 2023.
- Schor, Juliet B. and Samuel Bowles**, “Employment Rents and the Incidence of Strikes,” *The Review of Economics and Statistics*, 1987, *69* (4), 584–592.

Selyukh, Alina, “Retailers Howled About Theft Last Year. Why Not Now?,” NPR, March 11 2024.

Shapiro, Carl and Joseph E. Stiglitz, “Equilibrium Unemployment as a Worker Discipline Device,” *American Economic Review*, 1984, 74 (3), 433–444.

Tyco Retail Solutions and PlanetRetail RNG, “2018 Sensormatic Global Shrink Index,” Technical Report, Tyco Retail Solutions 2018. Global shrink of 1.82 percent of sales, approximately \$100 billion, across 14 countries.